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Benchmark Reliability for Multilingual LLM Performance Prediction: Extending the Signal-and-Noise Framework

Author:

Eléonore Hasler, École Polytechnique

Advisor:

Antoine Bosselut, École Polytechnique fédérale de Lausanne

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Abstract

Because training large language models is expensive, researchers often experiment on small models with different training corpora and, from the ranking of the models' scores on benchmarks, predict the ranking of larger versions of those models. This allows them to choose the optimal training mixture for the final large model at a low cost.
In this study, we build on the Signal-and-Noise framework introduced by Heineman et al. [10], which measures how reliably a benchmark preserve the ranking of models of different training data mixtures. We propose a new noise measure to estimate the reliability of a benchmark for such predictions. Most importantly, we extend this framework to multilingual benchmarks, which have largely been overlooked for such analyses, by analyzing properties of downstream multilingual tasks, as well as large evaluation sets such as Flores+ and Wiki40B.

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1 Introduction

1.1 Motivation

The importance of the choice of training dataset has been shown to be crucial for increasing models' performance, especially multilingual ones. However, the extreme demand for high-quality hardware and the extensive training time makes developing such models costly. Researchers therefore need a way to make informed decisions on model pre-training data mixtures before committing to a large-scale training run. Nevertheless, such decisions are hard to make without knowing in advance how much a choice in training data will influence the final model's performance.

One way to address this challenge is to train smaller proxy models, estimate how their performance scales to larger models, and use those predictions to guide design decisions. This approach relies critically on one assumption: that the benchmarks used to evaluate small models are reliable enough that the ranking of models at a small scale will reflect their ranking at a larger scale. If this assumption breaks down, decisions made at small scale may lead researchers to take wrong decisions on the pre-training data mixture, wasting enormous amounts of compute and resources.

This reliability assumption becomes especially fragile in multilingual settings. Most benchmark development and evaluation methodology has been driven by English-first research, and the tools developed to assess benchmark reliability have been validated almost exclusively on English tasks. Yet multilingual models face fundamentally different challenges: low-resource languages are underrepresented in training data, small models may perform near randomly on harder multilingual tasks, and the linguistic diversity of evaluation sets introduces sources of variability that English-centric frameworks were never designed to handle.

Previous research by Heineman et al. [10] provides a foundation for measuring how reliably benchmarks differentiate between models and how stable those measurements are across runs. This framework has shown promise for English benchmarks, but has never been tested in multilingual settings. Given the growing importance of multilingual NLP and the cost of training large multilingual models, understanding whether existing reliability tools extend to this setting — or whether they fail in predictable ways — is both urgent and crucial.

In this work, we focus on multilingual benchmarks and ask: *Can we find properties that will tell us to which extent a multilingual benchmark will reliably provide a ranking between small models that will hold at a larger scale?*

1.2 Contribution

This study is based on the Signal and Noise framework proposed by Heineman et al. [10]. The details of this work are explained in Section 2. Our work can be summarised as follows:

1. We verify that the method they propose for assessing benchmark reliability produces consistent results when reproduced, establishing a framework we can build on.

2. We propose a more practical and effective way to measure benchmark reliability by introducing *benchmark noise*.

3. We test whether the reliability of small-scale evaluations as predictors of large-scale model performance holds for multilingual benchmarks, and find that it is weaker than in English-only settings. We hypothesise this result to be mostly due to the English-first nature of the small models used.

4. We investigate whether evaluating model performance on raw text corpora using a continuous metric leads to more reliable predictions in the multilingual setting.

2 Background

2.1 Importance of training data mixture in model's performance

When building a language model, one can choose to make changes to the architecture such as varying the attention mechanism, the activation function, the tokenizer, or the number of layers in order to improve performance on downstream tasks. However, recent work has shown that improvements in data quality and mixing is a crucial component change in hyperparameter tuning in determining model performance.

Liu et al. [13] show that data ratios act as the primary driver of performance on downstream benchmarks. They infer those results from comparing the performance of 64 models, each with 1B parameters, trained on different data mixtures. They also highlight the need for automatic approaches to predict the effect of training data mixture in model's performance, as the effect of data mixture on performance is often counterintuitive — the best-performing mixtures frequently contradict what domain expertise or common sense would suggest. This further motivates the framework we develop in this work. Additionally, Ye et al. [21] establish that data domain proportions follow predictable mixing laws, making data composition a structured and optimisable design variable rather than an arbitrary choice. Using these laws to determine the optimal mixture, a 1B model trained on 100B tokens from RedPajama reaches performance comparable to a model trained on the default mixture for 48% more steps. This experiment demonstrates that principled data composition can substitute for a substantial increase in compute.

Wang et al. [20] survey current practices in LLM pretraining data management and highlight that domain composition, data quality, and filtering strategies are among the primary factors determining model performance, with the optimal data mixture being difficult to determine without systematic experimentation.

In addition, multilingual models face particular challenges in the choice of training data, making the prediction of the influence of a training data mixture on downstream performance non-trivial. This makes our work particularly relevant, as we provide a method to predict model performance based on different choices of training data.

2.2 Instability of Model Scores

Confirming the validity of scores obtained when evaluating language models is particularly important, but also particularly difficult. Language models are by definition non-deterministic: evaluating a model on a given benchmark multiple times will yield different outputs. It is important to account for this noise due to the stochastic nature of language models — referred to as *modeling noise* — when attesting to the validity of rankings between models evaluated on downstream tasks. On the other hand, benchmarks are also subject to instability, since they rely on static and often contaminated data, and their scores can fluctuate drastically based on minor changes in prompt formatting, the selection of few-shot examples, or even the order of answer choices.

2.2.1 Modeling Noise

When using small models' scores to predict the performance of larger models, we need to account for the instability of evaluation scores. The noise due to the model itself is what we call modeling noise. Madaan et al. [14] were among the first to quantify this noise systematically in the context of LLM evaluation. They measure the stability of model improvement (i.e., the monotonicity of the model) and the noise (i.e., the variation in scores due to the choice of initialization seeds). While this provides a meaningful measure of modeling noise, it requires training several models under the same configuration, which is expensive.

Heineman et al. [10] build on this by showing that checkpoint-to-checkpoint noise — the variability of a single model's score across its final training steps — is highly correlated with seed noise ($R^2 = 0.82$) and data order noise ($R^2 = 0.86$), and can therefore serve as a cheap substitute that requires no additional training runs.

They formalise this into a complete framework for measuring benchmark reliability. They define *signal* as the relative dispersion of scores across a population of models trained on different data recipes:

$$\mathrm{Signal}(M) = \mathrm{Rel.\,Dispersion}(M) = \frac{\max(|m_j - m_k|)}{\bar{m}} \tag{1}$$

and *noise* as the relative standard deviation of a single model's scores across its final n training checkpoints:

$$\mathrm{Noise}(m) = \mathrm{Rel.\,Std.}(m) = \frac{1}{\bar{m}}\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(m_i - \bar{m})^2} \tag{2}$$

In short, the signal reflects the ability of the benchmark to discriminate between models, and the noise reflects how stable the score of a given model is. We use this definition of signal throughout our experiments and use their checkpoint-to-checkpoint noise metric to sanity-check their results.

2.2.2 Intrinsic Noise of Benchmarks

For a long time, benchmarks have been used as ground truth to attest to a model's abilities. However, recent work has challenged this view, showing that benchmarks are unstable and that a margin of error should be considered due to their inherent instability.

Brown et al. [4] highlight that performance is highly sensitive to the selection and order of few-shot examples, with variations of 10–15% in classification tasks due to prompt permutation, alongside issues with data contamination. Subsequent work further demonstrates that small changes in prompt formatting, the distinction between log-likelihood and generative output, and data leakage create significant performance fluctuations, suggesting that benchmark gaps of 2–3% are often merely artefacts of formatting or training data proximity.

Card et al. [5] invalidate many claims of model improvement reported in NLP papers, demonstrating that those improvements are not statistically significant. They show that the vast majority of commonly used NLP benchmarks have insufficient statistical power to reliably distinguish between models whose true performance is close.

Berg-Kirkpatrick et al. [2] and Dehghani et al. [8] both argue that benchmark scores are random variables rather than fixed quantities, because any benchmark is a finite sample of questions drawn from a larger population of possible questions. They show that many reported differences between models are not statistically significant once this uncertainty is properly quantified, and that the precision of a benchmark score depends heavily on the number and independence of its questions.

Together, these papers motivate our approach of accounting for noise due to the benchmark itself, establishing that benchmark scores carry inherent variance that must be accounted for.

2.3 Work on Performance Prediction

2.3.1 Decision Accuracy

If a small model trained on data mixture A outperforms one trained on mixture B, will this advantage persist at a larger scale? The ability to rely on small-scale rankings to make this claim would allow researchers to compare architectures, training methods, and data mixtures cheaply before committing to an expensive large-scale training run. This is what Heineman et al. [10] define as *decision accuracy*.

They demonstrate that the signal-to-noise ratio (SNR), defined as:

$$\mathrm{SNR} = \frac{\mathrm{Signal}}{\mathrm{Noise}} = \frac{\text{Relative dispersion (final checkpoints)}}{\text{Relative std. deviation (final } n \text{ checkpoints)}} \tag{3}$$

is strongly correlated with decision accuracy ($R = 0.791$, $R^2 = 0.626$). We therefore use SNR as a proxy for evaluating how much a ranking of small models on a benchmark agrees with the corresponding ranking of larger models.

2.3.2 Scaling Laws: Predicting Large Model Performance

The underlying assumption for predicting models' ranking is that small model behaviour extrapolates meaningfully to large models. Previous work has used scaling laws to predict downstream task performance by first predicting task loss,[1] then using the predicted loss to predict task performance (e.g., accuracy).

Predicting the task loss was formalised by Kaplan et al. [?], who showed that language model loss follows predictable power laws as a function of the number of model parameters N, training tokens D, and compute budget C:

$$L(N, D) = \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}} + E \tag{4}$$

where L is the language modelling loss and A, B, α, β, E are fitted constants.

Bhagia et al. [?] extend this to downstream task performance by adding a sigmoid function that maps predicted loss to predicted task accuracy:

$$U(L) = \frac{a}{1 + e^{-k(L - L_0)}} + b \tag{5}$$

where a and b are the upper and lower performance asymptotes,[2] k is the growth rate, and L_0 is the inflection point indicating the loss threshold at which the model begins to meaningfully solve the task.

The scaling law prediction error is then defined as:

$$\text{Prediction error} = \frac{|\text{Predicted value} - \text{True value}|}{|\text{True value}|} \tag{6}$$

Although we do not directly compute scaling law prediction errors in our study, these scaling laws form the theoretical foundation of our framework. Heineman et al. [10] show that checkpoint-to-checkpoint noise correlates strongly with prediction error. Intuitively, a benchmark on which model scores fluctuate heavily across training steps will produce an unreliable scaling law fit, and therefore a higher prediction error. This means that noise is a practical indicator of how well a benchmark can be used to predict large model performance. It follows that SNR, which combines noise with signal, is a principled metric for identifying benchmarks that are both stable enough and discriminative enough to support reliable performance prediction at scale.

2.4 Motivation for Using Continuous Metrics

Madaan et al. [14] recommend using continuous metrics over discrete ones such as accuracy when comparing models. For choice-based tasks, they use the probability mass of the predicted answer; for generation-based tasks, they use the negative log-likelihood of the target answer. Their findings suggest that continuous metrics are more reliable for distinguishing model improvements than discrete metrics. They also find that continuous metrics exhibit higher monotonicity during training in almost all cases, meaning model performance improves more stably over the course of pre-training when measured continuously. They conclude that continuous metrics are less confounded by noise and may therefore be more reliable for making decisions during model development.

Heineman et al. [10] reach the same conclusion, recommending bits-per-byte (BPB) as the continuous metric of choice. BPB measures how many bits of information the model requires on average to encode each byte of text, and serves as a language-model-native analogue of perplexity. Like probability mass, it sidesteps the threshold effects that make discrete accuracy noisy — a model that is nearly correct still receives partial

[1] The task loss refers to the average negative log-likelihood of the tokens in the correct answer given the preceding context. It measures how well the model predicts each token of the correct answer in sequence. A low loss means the model assigns high probability to the correct tokens, while a high loss means the model is uncertain about them.

[2] The upper performance asymptote corresponds to the maximum theoretically achievable task performance as loss approaches zero (i.e., a perfect model); the lower asymptote corresponds to when loss approaches infinity (i.e., a model that does not learn).

credit in the loss, rather than being counted as a full error. Across the benchmarks they study, switching from accuracy to BPB consistently raises SNR and improves decision accuracy, suggesting that BPB is also better suited for performance prediction. This motivates our choice to extend our work by adopting BPB metrics to evaluate models on raw text corpora.

3 Methodology

We build on the Signal and Noise framework of Heineman et al. [10] to evaluate the reliability of benchmarks for performance prediction. We first reproduce their analysis on English benchmarks, then propose a stronger noise metric, and finally investigate whether the framework yields meaningful results on multilingual benchmarks. This section defines the metrics and evaluation protocol used across all experiments.

3.1 Prediction Metrics

Signal. Following Heineman et al. [10], we quantify the *signal* of a benchmark by the relative dispersion of model scores across training-data mixtures, for a fixed benchmark and model size:

$$\text{Signal}(M) = \frac{\max_{m \in M} s(m) - \min_{m \in M} s(m)}{\frac{1}{|M|} \sum_{m \in M} s(m)} \tag{7}$$

where $s(m)$ denotes the score of model m on the benchmark, and M is the set of models being compared. A higher signal means that the benchmark more clearly separates different training-data mixtures.

Checkpoint-to-checkpoint noise. In the reproduction experiment, we use the same checkpoint-based notion of noise as in the original signal-and-noise framework: model scores are tracked over training, and variability near the end of training is used as a measure of instability. Concretely, for each model m, we compute the standard deviation of its scores over the final portion of the training curve,

$$\sigma_{\text{step}}(m) = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \big(m_i - \bar{m}\big)^2} \tag{8}$$

where $m_1, \ldots, m_n$ are the selected late-training checkpoint scores and $\bar{m}$ is their mean. In our implementation, this standard deviation is computed over the last 30% of available checkpoints. These per-model values are then aggregated across models and normalized by the benchmark-level mean score, yielding a benchmark-level checkpoint noise estimate:

$$\text{Checkpoint noise}(M) = \frac{\frac{1}{|M|} \sum_{m \in M} \sigma_{\text{step}}(m)}{\mu(M)} \tag{9}$$

where $\mu(M)$ denotes the mean benchmark score across the models in M. This metric is used in the reproduction experiment to compare our results to the original paper. Its main limitation is that it requires access to intermediate training checkpoints, which are often unavailable for public models.

Benchmark noise. We also introduce an alternative noise metric based on score variation across folds of the evaluation benchmark rather than across training checkpoints. The goal is the same: to estimate how stable benchmark scores are under repeated measurement.

The exact procedure differs slightly between the two experiments. In the reproduction experiment, each benchmark is shuffled once with a fixed seed, partitioned into $k = 5$ folds, and each model is evaluated independently on every fold. In the multilingual experiment, the model is evaluated once on the full benchmark, per-example outputs are stored, and fold scores are obtained afterward by partitioning the stored predictions into $k = 5$ folds without rerunning inference.

For a given model m, let $m_1, \ldots, m_k$ denote its fold-level scores. Benchmark noise is defined as the relative standard deviation across these fold scores:

$$\text{Noise}_{\text{bench}}(m) = \frac{\sqrt{\frac{1}{k}\sum_{i=1}^{k}\left(m_i - \bar{m}\right)^2}}{\bar{m}} \tag{10}$$

where $\bar{m} = \frac{1}{k}\sum_{i=1}^{k} m_i$. We then average this quantity across models:

$$\text{Benchmark noise}(M) = \frac{1}{|M|}\sum_{m \in M} \text{Noise}_{\text{bench}}(m) \tag{11}$$

In the multilingual setting, fold scores for multi-subtask benchmarks are first aggregated within each fold using the number of evaluation instances as weights. Benchmark noise is appealing because, unlike checkpoint-to-checkpoint noise, it does not require access to intermediate checkpoints.

Signal-to-noise ratio. The signal-to-noise ratio (SNR) combines the benchmark's discriminative power and its instability into a single quantity:

$$\text{SNR}(M) = \frac{\text{Signal}(M)}{\text{Noise}(M)} \tag{12}$$

In our experiments, we instantiate this definition with either checkpoint-to-checkpoint noise or benchmark noise. A higher SNR indicates that score differences across models are large relative to the benchmark's intrinsic variability.

Decision accuracy. Decision accuracy measures how well the ranking induced by smaller proxy models matches the ranking induced by larger target models. It is defined as the fraction of pairwise comparisons for which the two rankings agree:

$$\text{Decisionaccuracy} = \frac{1}{|\mathcal{P}|}\sum_{(a,b)\in\mathcal{P}} \mathbf{1}\left[\left(\text{rank}_{1\text{B}}(a) < \text{rank}_{1\text{B}}(b)\right) \iff \left(\text{rank}_{s}(a) < \text{rank}_{s}(b)\right)\right] \tag{13}$$

where $\mathcal{P}$ is the set of unordered pairs of training-data mixtures that appear in both rankings, $\text{rank}_{1\text{B}}$ is the ordering induced by the 1B models, and rank_s is the ordering induced by models of size s.

Heineman et al. [10] show that benchmarks with higher SNR tend to yield higher decision accuracy. We therefore use SNR as our main proxy for benchmark reliability, and evaluate it by measuring its correlation with decision accuracy.

3.2 Performance Metrics

Accuracy. For our evaluation on downstream tasks — including the Signal and Noise replication and the multilingual tasks from the LM Evaluation Harness — we use accuracy as the primary metric.

Bits-per-byte. When evaluating models on large text evaluation sets, we use BPB, defined as the average number of bits required by the model to encode each byte of text. Unlike accuracy, which is discrete and requires a "correct" label, BPB is a continuous, threshold-free alternative to accuracy and is better suited for evaluating models on raw corpora such as Flores+ [16] and Wiki40B [9].

Heineman et al. [10] define BPB specifically for downstream completion tasks. In their framework, BPB is calculated only on the correct continuation (target) given a fixed prompt:

$$\text{BPB}_{\text{task}} = \frac{-\log_2 P(\text{target} \mid \text{prompt})}{\text{Bytes}(\text{target})} \tag{14}$$

In contrast, our work utilizes a *rolling* BPB approach suitable for raw text. Here, the model processes the entire text autoregressively, where each token becomes part of the context for the next, and the loss is normalized by the total byte length of the UTF-8 string:

$$\text{BPB}_{\text{rolling}} = \frac{\sum_{i=1}^{n} -\log_2 P(x_i \mid x_{<i})}{\text{Bytes}(\text{Total Text})} \tag{15}$$

Our rolling definition is significantly more robust for multilingual tasks. Downstream task BPB (Eq. 14) is often confounded by "instruction-following" noise; if a model does not understand an English-centric prompt, its score on a multilingual task may appear poor regardless of its actual proficiency in the target language. By using a rolling BPB (Eq. 15) on raw corpora, we evaluate the model's ability to model the language directly, bypassing the bottleneck of prompt comprehension which is particularly prevalent in low-resource settings.

3.3 Models

We use the DataDecide models introduced by Magnusson et al. [?]. These training mixtures are primarily English-centric, high-quality pre-training corpora derived from massive web crawls (Common Crawl) commonly used in the training data mixtures of large language models.

Reproduction experiment. For the replication of the Signal and Noise experiments, we use the DataDecide models available in allenai/signal-and-noise, covering the following training-mixture families:

```
{DCLM-baseline | c4 | dclm_{ft7percentile_fw{2|3} | fw_top{3|10}} |
dolma_{v1-6-and-sources-baseline | 17(-{25p-DCLM-baseline-75p |
50p-DCLM-baseline-50p | 75p-DCLM-baseline-25p})?} |
falcon(_and_cc(_{eli5_oh_top{10p|20p} | og_eli5_oh_top10p |
tulu_qc_top10})?)? | fineweb_edu_dedup | prox_fineweb_pro |
no_{code | flan | math_no_code | reddit} |
pos_eli5_oh_neg_dclm_refinedweb_steps_2000_lr3e4_top{10p|20p}}
```

Concretely, this dataset includes model sizes `{4M, 6M, 8M, 10M, 14M, 16M, 20M, 60M, 90M, 150M, 300M, 530M, 750M, 1B}` and seeds `{14, 15, 6198}`, with seed and checkpoint coverage varying across mix/size combinations.

Benchmark noise experiment. To support the definition of the new noise measure, the *benchmark noise*, we use a subset of training mixtures with a single fixed seed (6198) and one fixed training step per model:

```
allenai/DataDecide-(dclm-baseline-20M | falcon-and-cc-qc-orig-10p-530M |
dclm-baseline-{25p|50p|75p}-dolma1.7-{75p|50p|25p}-{150M|530M|1B} |
dclm-baseline-qc-{7p-fw2|7p-fw3|fw-3p|fw-10p}-{150M|530M|1B} |
dolma1_6plus-{150M|530M|1B})
```

We therefore assume that score obtained from a single run of a model's final checkpoint is sufficiently stable.

The models used to compute benchmark noise span sizes 20M, 150M, 530M, and 1B, selected to cover different training corpora and thereby mitigate the risk of findings holding only for a specific dataset. We started testing our setup using `dclm-baseline-20M` and `falcon-and-cc-qc-orig-10p-530M` models, we added them to the list as well. For this experiment, the signal and decision accuracy are computed identically to the reproduction experiment described above.

Multilingual experiment. For the multilingual analysis, we evaluate DataDecide checkpoints on downstream multilingual tasks and on subsets of BPB evaluation sets Flores+ and Wiki40B. Similarly to the Benchmark noise experiment described above, we assume that the scores of models' last checkpoint are stable enough. The evaluated models are the following:

```
datadecide-(
dclm-baseline(-{25p-dolma1-7-75p | 50p-dolma1-7-50p | 75p-dolma1-7-25p})?-
{4m|20m|60m|90m|150m|300m|530m|750m|1b} |
dclm-baseline-qc-{10p|20p|7p-fw2|7p-fw3|fw-3p|fw-10p}-
{4m|20m|60m|90m|150m|300m|530m|750m|1b} |
dolma1-6plus-{4m|20m|60m|90m|150m|300m|530m|750m|1b}
)
```

Full details on the composition of training mixtures and model parameter counts are provided in Appendix A.

3.4 Benchmarks

Reproduction benchmarks. The benchmarks used to replicate the Signal and Noise paper results are English multiple-choice and completion tasks evaluating commonsense reasoning, scientific knowledge, and general world understanding. The benchmarks are MMLU [11], ARC [7], BoolQ [6], CSQA [19], OBQA [15], PiQA [3], SocialIQA [18], HellaSwag [22], WinoGrande [17]. The details on the format, domains evaluated and number instances are reported in Table 3.

Tasks used for benchmark noise experiment. This experiment uses the Reproduction benchmarks.

Downstream multilingual tasks. For the extension to the multilingual setting, we use a selection of lm-evaluation-harness repository multilingual tasks, evaluated using accuracy. We chose tasks that collectively cover a wide range of languages, including both high-resource and low-resource ones; details can be found in Table 4.

Bits-per-byte evaluation sets. For evaluation of multilingual text corpora, we created disjoint subsets from both Flores+ [16] and Wiki40B [9] evaluated using the BPB metric: `flores_plus_bpb_1` to `flores_plus_bpb_10` and `wiki40b_bpb_1` to `wiki40b_bpb_5`. These are raw datasets covering a wide range of languages; details on the languages covered can be found in Table 5

4 Reproduction of Existing Signal-and-Noise Analysis

To begin our study, we build on the Signal and Noise framework of Heineman et al. [10]. We therefore start by reproducing their analysis using their pre-computed model scores.

4.1 Setup for Reproduction Experiment

We use these full DataDecide model suite with seed and training step variations defined in 3.3. We compute checkpoint-to-checkpoint noise from the final 30% of training steps. The target model size whose performance we aim to predict is 1B. We compute the score used for signal and decision accuracy by averaging the last 10% of checkpoints of each model to obtain more stable scores. The benchmarks used for this reproduction experiments are detailed in 3.4.

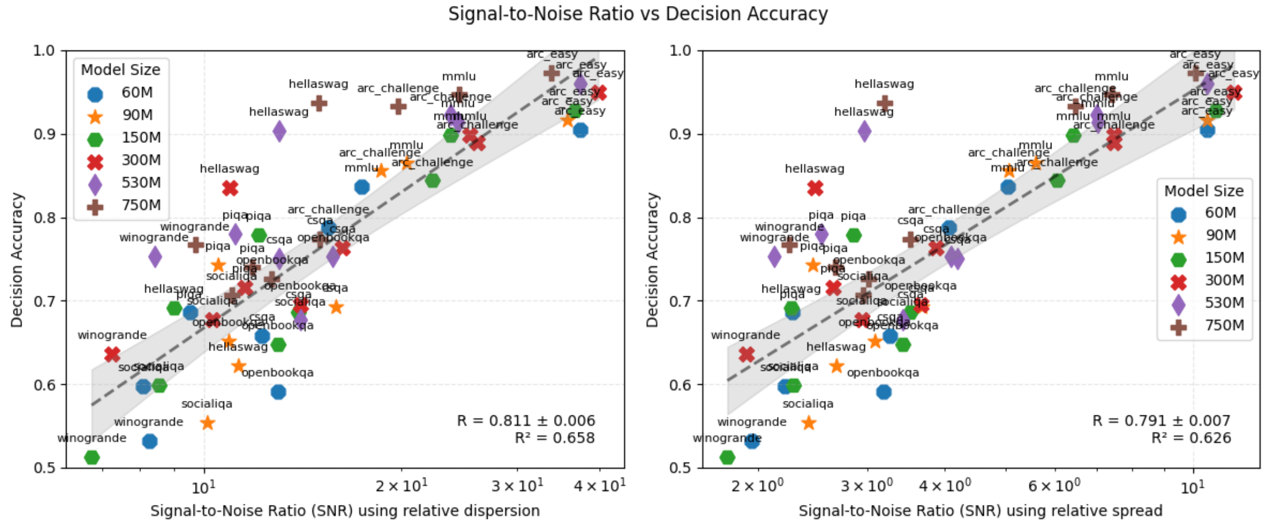

Figure 1: Signal-to-noise ratio vs. decision accuracy comparing both measures of signal, relative spread and relative dispersion. Using relative dispersion as the signal metric gives a stronger correlation between SNR and decision accuracy. It is therefore a more suitable metric for evaluating how well a benchmark discriminates between models, as it better reflects the ability of the benchmark to preserve ranking across model sizes.

4.2 Validation of the Relative Dispersion as a Signal Measure

In their study, Heineman et al.[10] propose two measures of signal: relative dispersion (defined in Section 3) and relative standard deviation, referred to as relative spread. While they claim that relative dispersion is a better signal metric than relative spread, their published plots use the latter. Our re-analysis confirms their claim: as shown in Figure 1, using relative dispersion as the signal metric yields a stronger correlation between SNR and decision accuracy ($R = 0.811$, $R^2 = 0.658$) compared to relative spread ($R = 0.791$, $R^2 = 0.626$). Relative dispersion is therefore a more appropriate metric for evaluating how well a benchmark discriminates between models, as it better reflects the benchmark's ability to preserve ranking across model sizes. We adopt relative dispersion as our signal metric for all subsequent experiments.

While our analysis supports the validity of the framework for English-centric models and benchmarks, the noise metric still requires access to a high level of granularity in the models. In particular, it requires access to a relatively large number of training checkpoints for all small proxy models. Since the DataDecide suite is the only open model family with such granularity, this constrains the analysis to their models. We therefore propose a new noise metric that does not require access to intermediate training checkpoints, making the framework considerably easier to apply in practice.

5 Benchmark Noise as a Stronger Noise Metric for SNR

We propose to evaluate noise as the instability arising from the benchmark itself. Our primary motivation comes from prior studies [5], which establish that benchmark scores are noisy estimates of model performance, partly because a benchmark is by definition composed of a finite and potentially unrepresentative sample of questions.

The second motivation is practical. While checkpoint-to-checkpoint noise requires access to intermediate training checkpoints — which are rarely publicly available — the k-fold benchmark noise is computable from a single evaluation run on any model, provided that per-question results are stored.

Our hypothesis is that both metrics capture the inherent instability of a benchmark's scores. A benchmark will by nature produce scores that fluctuate, regardless of whether that fluctuation is measured across training checkpoints or across folds of the evaluation set. This shared target is what leads us to expect a correlation between the two metrics, despite them measuring score instability from different sources. Additionally, since both noise measures are formulated as relative standard deviations, they are directly comparable on the same scale without any additional normalization, making the correlation straightforward to interpret.

5.1 Setup for the Benchmark Noise Experiment

This experiment uses the same reproduction benchmarks described in Section 3.4, the same checkpoint-to-checkpoint noise values computed in Section 4 and the full DataDecide model suite with seed and training step variations across sizes from 10M to 750M.

To compute benchmark noise, we use the Datadecide models defined in Section 3.3: models of sizes 20M, 150M, 530M, and 1B, spanning ten different training corpora.

Unlike checkpoint-to-checkpoint noise, which requires access to many intermediate training checkpoints across multiple seeds, benchmark noise is computed from a single evaluation run of each model's final checkpoint. In this experiment, per-question results are stored during evaluation, and the $k = 5$ fold scores are then derived by partitioning those stored predictions without rerunning inference. This makes our pipeline adaptable for changing the value of k and avoids redundant compute by reusing stored evaluation outputs.

Signal and decision accuracy are computed identically to the reproduction experiment: signal is the relative dispersion of scores across training mixtures, and decision accuracy measures agreement between small model rankings and 1B model rankings.

5.2 Correlation with Checkpoint-to-Checkpoint Noise

Motivation of Using Small Models to Estimate Benchmark Noise. For conciseness, we present in Figure 2 only the results using benchmark noise computed from 150M models. This choice is central to our goal: we want the SNR to be a reliable predictor of the ranking of 1B models from significantly smaller proxy models. If we needed to rely on 530M or 1B models to compute a reliable noise estimate, the framework would lose much of its practical value — the whole point is to avoid running large models. We therefore need benchmark noise to be reliable even when computed from small models only, and our result shows that the approximation `Benchmark noise` $\approx$ `Checkpoint-to-checkpoint noise` holds even at that scale. We also verify that this relation does not break down as models grow larger: using 300M, 530M and 1B models to compute benchmark noise yields the same conclusion, confirming that the result is robust across model sizes and not an artefact of the small scale. These additional results are provided in Appendix A.

Empirical Correlation Between Benchmark and Checkpoint Noise. Evaluating on six English downstream tasks, we find a strong correlation between checkpoint-to-checkpoint noise and benchmark noise ($R = 0.714$, $R^2 = 0.509$), as shown in Figure 2. This confirms that benchmark noise is a good substitute for checkpoint-to-checkpoint noise. The correlation is not perfect, which is expected since the two metrics measure instability from fundamentally different sources: one from the stochasticity of the training process, the other from the finite sampling of benchmark questions. What matters is that they are close enough to be interchangeable for the purpose of SNR computation. Notably, this correlation is obtained without any calibration or rescaling between the two metrics, which is possible because both are formulated as relative standard deviations. We therefore adopt benchmark noise as our noise metric for all subsequent experiments, replacing checkpoint-to-checkpoint noise entirely.

5.3 Improvement in SNR Predictive Power

Now that benchmark noise has been shown to correlate strongly with checkpoint-to-checkpoint noise, we use it in our signal-to-noise ratio computation on downstream tasks. The result, shown in Figure 3, is a marked

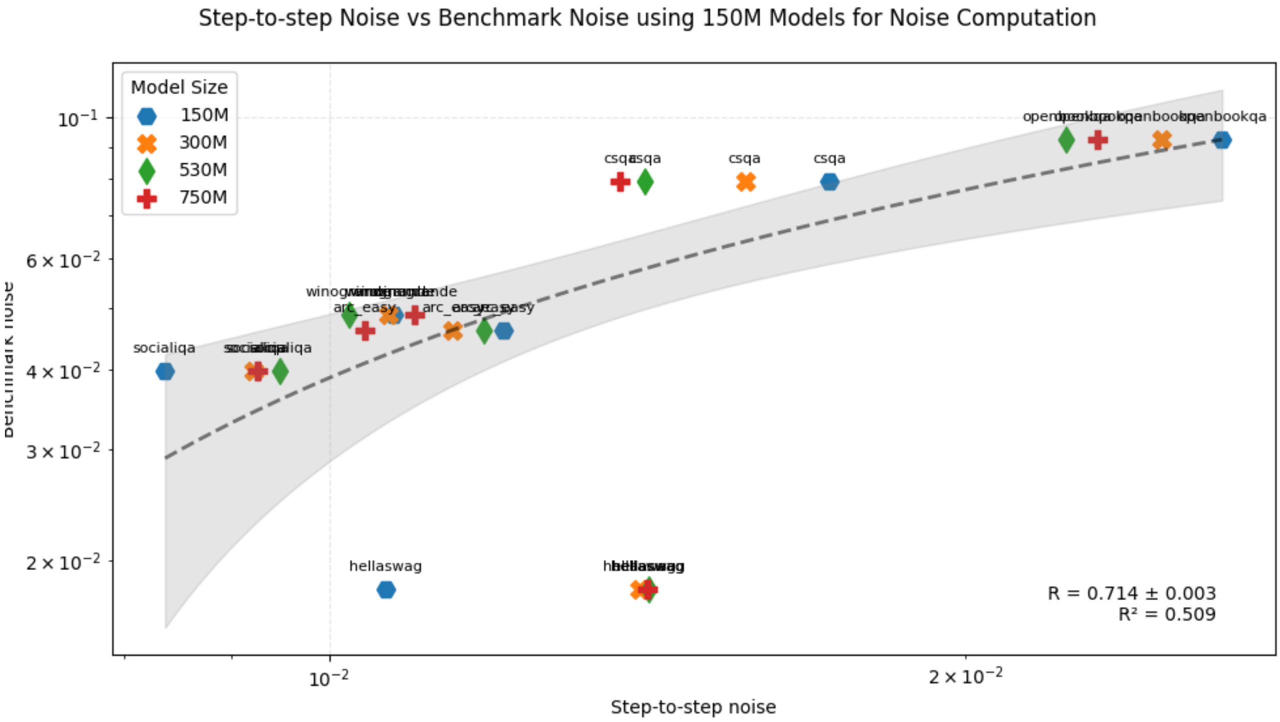

Figure 2: Step-to-step noise vs. benchmark noise. This figure compares the correlation between checkpoint-to-checkpoint noise and k-fold benchmark noise. It uses DataDecide models spanning 150M to 750M parameters. The models used to compute benchmark noise are all 150M models. Additional runs using models from 20M to 1B to compute benchmark noise have also shown a strong correlation between the two types of noise. Details can be found in Appendix A.

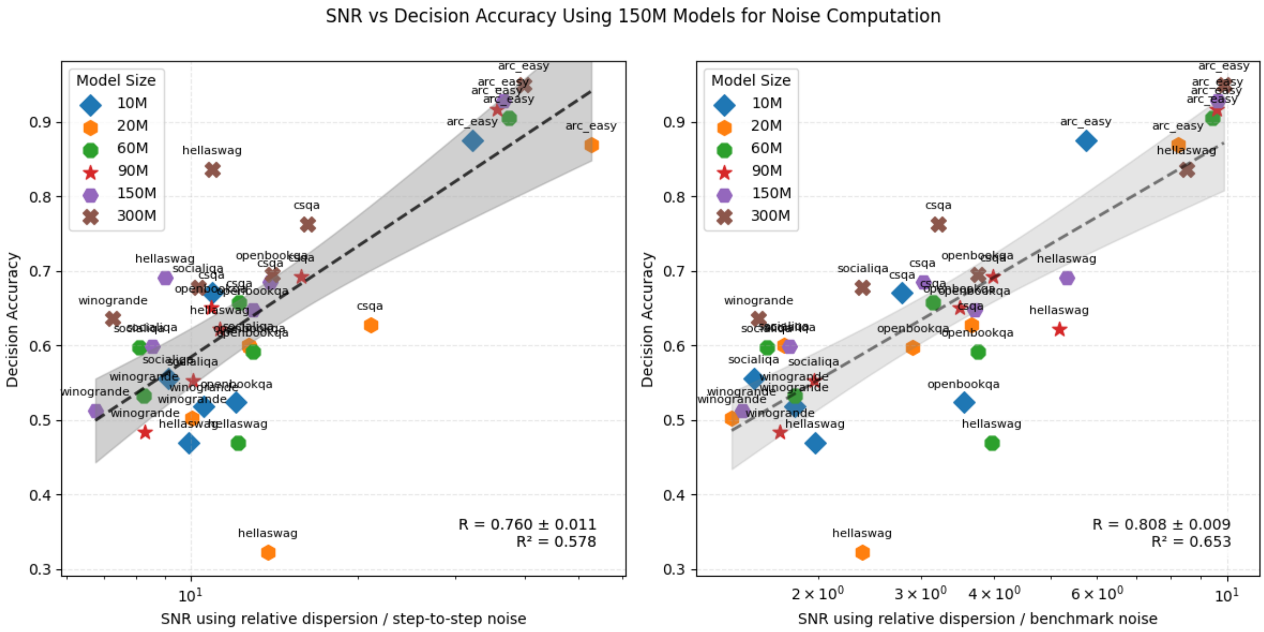

Figure 3: SNR vs. decision accuracy from small models, using 150M models for benchmark noise computation. The strong correlation demonstrates that benchmark noise is a good approximation for checkpoint-to-checkpoint noise, making it our new noise measure for all subsequent signal-to-noise ratio computations.

improvement in correlation with decision accuracy when using benchmark noise compared to checkpoint-to-checkpoint noise ($R = 0.808$, $R^2 = 0.653$).

Benchmark noise increases SNR reliability. This improvement is the central finding of our work. Recall that decision accuracy measures how often the ranking of small models on a benchmark agrees with the ranking of larger models — it is the quantity we ultimately care about when using small proxy models to guide pre-training decisions. The fact that replacing checkpoint-to-checkpoint noise with benchmark noise in the SNR formula improves the correlation with decision accuracy, and not just the correlation between the two noise metrics themselves, means that benchmark noise is not merely a convenient approximation — it is a *better* measure for the purpose of predicting which benchmarks will yield reliable rankings. This is a stronger claim than showing the two noise metrics are interchangeable, and it is what justifies benchmark noise as the preferred metric going forward.

Convenience of benchmark noise for our experiments. It is worth noting that this improvement comes despite benchmark noise being strictly easier to compute. Checkpoint-to-checkpoint noise requires access to many intermediate training checkpoints across multiple seeds, which is only possible for the DataDecide model suite. Benchmark noise requires a single evaluation run on any model, provided per-question results are stored. The fact that the easier-to-compute metric also performs better on the downstream criterion is a particularly encouraging result.

Using benchmark noise as a noise estimate is particularly well-suited to our goal of extrapolating the performance of small models. Indeed, the training checkpoints of very small models may not reflect a strong learning trend, since their performance is still highly variable. We therefore hypothesize that relying on the noise arising from the benchmark may be more reliable than relying on the not necessarily informative variations of training checkpoints for very small models.

The correlation found in Figure 3 is particularly compelling since it uses benchmark noise computed from

150M models, significantly smaller than the target size, and model scores from sizes as small as 10M to 300M. It shows that benchmark noise is, even at a very small scale, a good estimate of benchmark instability.

This justifies our metric choices for all subsequent analyses. All following computations of signal-to-noise ratio will use relative dispersion for signal and k-fold benchmark noise for noise.

6 Extension to Multilingual Downstream Tasks

Multilingual benchmarks have mostly been left out of signal-and-noise analyses. However, choosing multilingual training data may be even more important and a greater challenge than choosing English training data, given the under-representation of low-resource languages and the difficulty of cross-lingual transfer learning. It is therefore crucial to verify that we can reliably predict the performance of large models from smaller proxy models in a multilingual setting. To address this, we extend the Signal and Noise framework to multilingual downstream tasks and examine whether the results hold.

6.1 Experimental Setup

We evaluate all 99 DataDecide models (11 different training corpora $\times$ 9 sizes) on 29 multilingual tasks described in Section 3.4, covering both English-only and non-English tasks.

Figure 4 focuses on a subset of seven tasks, namely `bangla_mmlu`, `egyhellaswag`, `truthfulqa-multi`, `xwinograd`, `xstorycloze`, `belebele`, and `click`, selected because four of them evaluate at least five different languages each, and the remaining three (`bangla_mmlu`, `egyhellaswag`, and `click`) cover languages not present in the other benchmarks, namely Bengali, Egyptian Arabic, and Korean. The SNR vs. decision accuracy analysis in Figures 5, 6 and 7 uses all 29 tasks.

6.2 Near-Random Performance Explains the Most Unstable Benchmarks

Overall correlation. Evaluating all 99 DataDecide models, we obtain a correlation of $R = -0.088$ between SNR and decision accuracy across all seven multilingual tasks, with `click`, `bangla_mmlu`, and `belebele` being particularly anomalous. Their scores are highly unstable, with larger models not consistently achieving better results. Removing these three tasks raises the correlation to $R = 0.572$.

Explanation. The three most unstable tasks — `bangla_mmlu`, `egyhellaswag`, and `belebele` — share a common explanation: they cover languages that are severely underrepresented in the English-first DataDecide pre-training corpora.

For `belebele`, this instability is well-supported by the literature. The original Belebele paper [1] shows that for approximately 25 languages, translated instructions were not well understood, with accuracy falling below random chance, and that performance correlates strongly with resource availability. The paper explicitly shows that English-centric models such as Llama-1 perform very differently across language families, with performance dropping substantially for low-resource languages. For very small English-first models such as the DataDecide suite, performance across Belebele's 122 languages is highly variable and often near-random, which explains the instability we observe. This is an expected consequence of evaluating English-first models on a massively multilingual benchmark.

For `bangla_mmlu` and `egyhellaswag`, Bengali and Egyptian Arabic are low-resource languages that are severely underrepresented in DataDecide pre-training corpora. Model scores on these tasks are therefore essentially random, producing the instability we observe. While we cannot find specific literature documenting known instability issues for these particular tasks with small English-first models, we hypothezise that the under-representation of those languages in the pre-training data from the Web used in the Datadecide models is the reason for the near-random scores we observe.

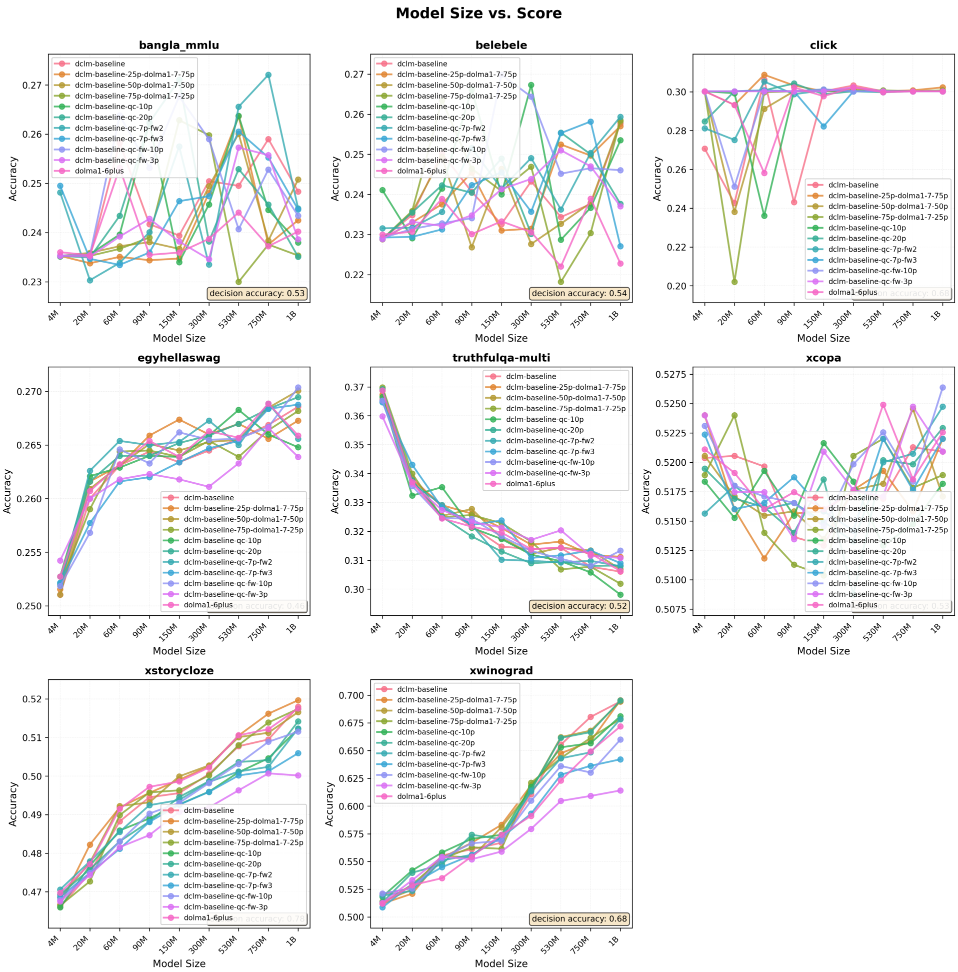

Figure 4: Model accuracy as a function of model size across multilingual downstream tasks, for all 11 DataDecide training corpora. The downward trend visible for `truthfulqa-multi` reflects the known *inverse scaling* phenomenon on this benchmark.

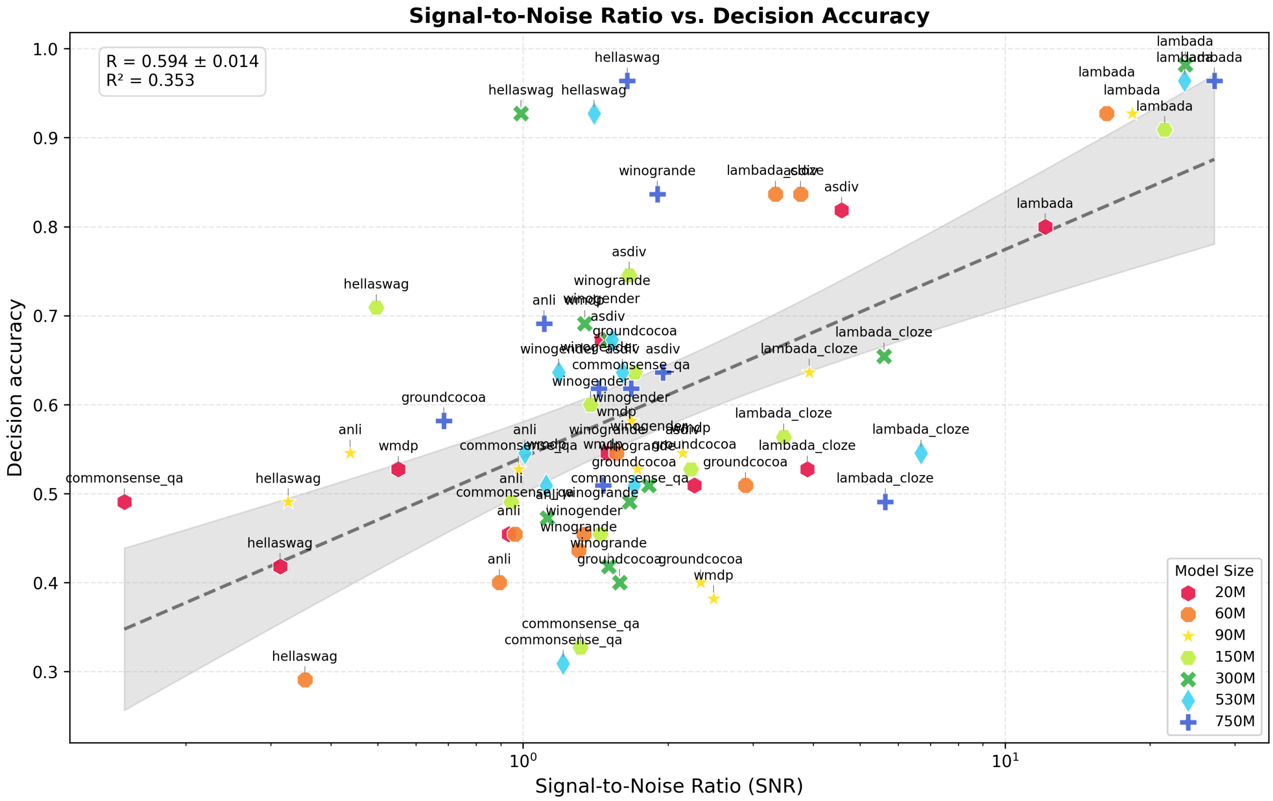

Figure 5: SNR vs. decision accuracy on English-only tasks ($R = 0.594$, $R^2 = 0.353$). The framework retains strong predictive power in the English setting, consistent with the results of the reproduction experiment.

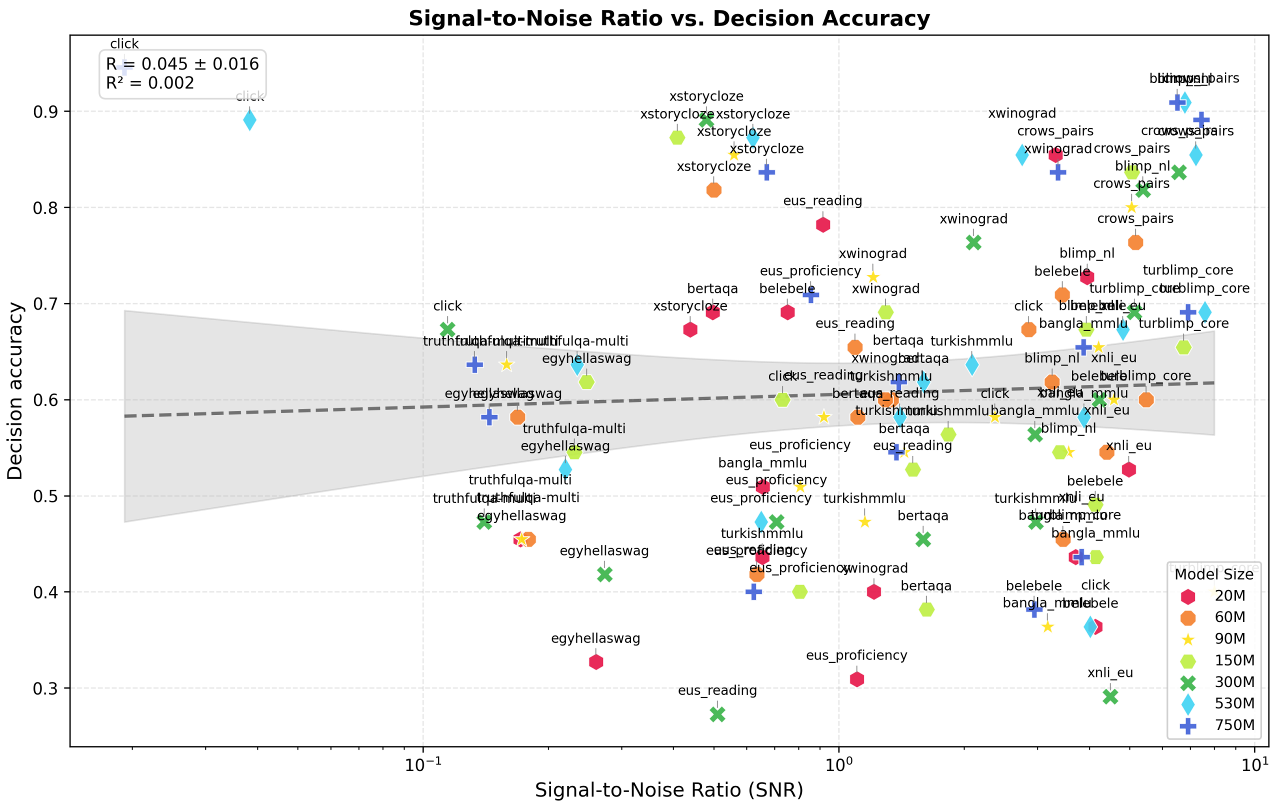

Figure 6: SNR vs. decision accuracy on all non-English only tasks ($R = 0.045$, $R^2 = 0.002$). The near-zero correlation shows that the framework does not reliably predict rankings in the multilingual setting when all tasks are included.

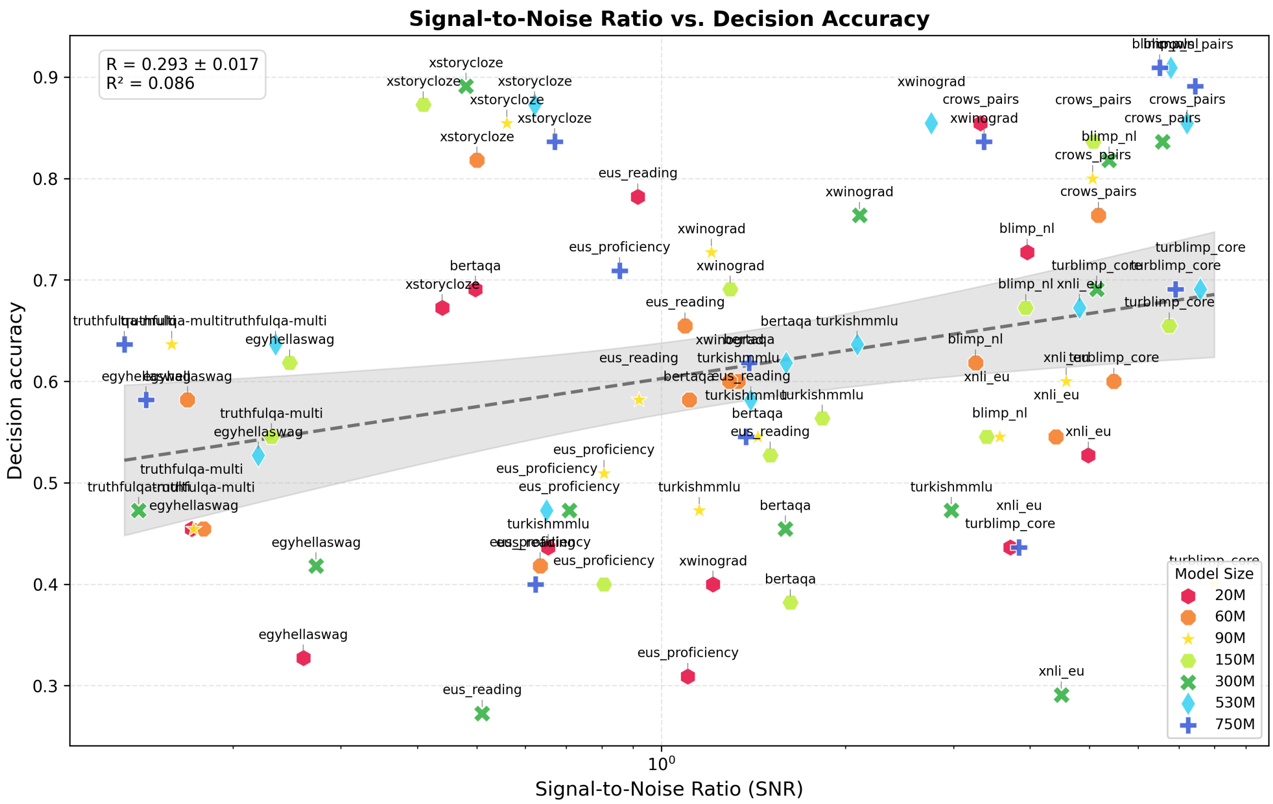

Figure 7: SNR vs. decision accuracy on non-English only tasks after removing the three most unstable benchmarks (`click`, `bangla_mmlu`, `belebele`) ($R = 0.293$, $R^2 = 0.086$). The partial recovery of the correlation suggests the framework retains some signal on tasks where models have meaningful linguistic competence.

6.3 Framework Reliability is Conditional on Model Competence

English-only vs. multilingual tasks. Figures 5, 6, and 7 show the SNR vs. decision accuracy plots separately for English-only and non-English tasks. On the English-only tasks, the framework yields a correlation of $R = 0.594$ ($R^2 = 0.353$). On the non-English tasks, the correlation drops to $R = 0.045$ ($R^2 = 0.002$) across all tasks, and rises to $R = 0.293$ ($R^2 = 0.086$) after removing the three most unstable benchmarks. The weaker correlation on multilingual tasks is consistent with the English-first nature of the DataDecide training corpora.

Explanation. These results suggest that the Signal and Noise framework holds for English tasks but weakens considerably in the multilingual setting when evaluated with English-first models. Two observations support this interpretation. First, the three tasks with the most unstable scores are precisely those covering languages most underrepresented in the training data. Second, removing those tasks raises the correlation from 0.045 to 0.293, suggesting that the framework retains some signal on multilingual tasks where the models have meaningful linguistic competence.

This does not necessarily mean the framework fails for multilingual settings in general. Rather, it suggests that the reliability of small-scale evaluations as predictors of large-scale performance is conditional on the proxy models having at least some competence in the evaluated languages. When models perform near-randomly on a task, the ranking they produce is uninformative, and no framework for benchmark reliability can recover meaningful signal from it. Extending this analysis to multilingual models trained on data that includes meaningful proportions of the evaluated languages remains an important direction for future work.

7 SNR Evaluation on Evaluation Sets Using BPB

So far in Sections 4–6 we rely on accuracy as the metric to evaluate models on downstream tasks. As discussed in Section 2, accuracy is a discrete metric, which means that a model that nearly predicts the correct answer receives the same score as one that is entirely wrong. Heineman et al. [10] and Madaan et al. [14] both recommend switching to continuous metrics, and in particular to bits-per-byte (BPB), to obtain more stable and discriminative benchmark scores. In this section, we investigate whether adopting rolling BPB on raw multilingual text corpora leads to more reliable SNR-based predictions than accuracy on downstream tasks.

7.1 Experimental Setup

Rather than evaluating models on structured question-answering tasks, we evaluate them directly on raw text corpora using rolling BPB (defined in Section 3). We use the same 99 DataDecide models as in Section 6 (11 training corpora $\times$ 9 sizes), evaluating only the final checkpoint of each model. We evaluate those models on the disjoint subsets of Flores+ and Wiki40B described in Section 3.4: `flores_plus_bpb_1` to `flores_plus_bpb_10` and `wiki40b_bpb_1` to `wiki40b_bpb_5`. Each subset covers a different combination of languages, and the BPB score of a model on a given subset is computed as the weighted average of the BPB scores obtained for each languages within the subset. Signal and benchmark noise are computed as defined in Section 3.1, using the same k-fold procedure as in the multilingual experiment.

7.2 High Decision Accuracy with Low Signal

Figure 8 shows models' BPB scores as a function of model size across all evaluation subsets. We make two main observations from those results. First, we notice that decision accuracy is consistently very high across all subsets, ranging from 0.77 to 0.96, with Wiki40B reaching 0.96. It shows that even small proxy models produce rankings that agree closely with 1B model rankings when evaluated on raw text corpora. This is a much stronger stronger result than the near-random decision accuracy observed on the multilingual downstream tasks in Section 6.

Second, the signal is relatively low across all subsets. Figure 9 shows that signal values hover between 0.05 and 0.16 across model sizes and evaluation sets. This is a direct consequence of the limited diversity of the DataDecide training mixtures: DCLM, Dolma, Falcon, FineWeb and their filtered derivatives all derive from Common Crawl and differ primarily in their filtering strategy rather than in their fundamental language or domain coverage. Since all models are therefore pre-trained on variations of the same English web crawl data, we can expect them to produce very similar BPB scores on multilingual corpora. The evaluation sets do not strongly discriminate between training mixtures because the mixtures themselves are not sufficiently diverse in their multilingual content to produce meaningfully different BPB values. Despite this, the rankings remain stable, as the consistently high decision accuracy reflects.

7.3 Using BPB Leads to Better Predictions

Figure 10 shows the SNR vs. decision accuracy plot for all BPB evaluation sets. The correlation is $R = 0.307$ ($R^2 = 0.094$), which is weak in absolute terms but substantially higher than the correlation obtained on non-English downstream tasks ($R = 0.045$, Section 6). This is a meaningful improvement, especially given that the models used are still English-first. It is due to this same limitation that the models produced near-random performance on many multilingual downstream tasks. Therefore, this strong increase in correlation supports the conclusion made by Heineman et al. [10] on English tasks: using BPB leads to better predictions of target-sized models' ranking.

The improvement over the downstream accuracy results can be attributed to two factors. First, rolling BPB bypasses the instruction-following bottleneck that makes accuracy unreliable for English-first models on non-English tasks: the model is evaluated on its ability to model the language directly, without needing to understand a prompt. Second, raw text corpora such as Flores+ and Wiki40B provide much larger and more consistent evaluation sets than downstream tasks, reducing per-language sample size variance.

The weak absolute correlation is largely explained by the low signal discussed above: when training mixtures are all variations of English web data, the BPB scores they produce on multilingual corpora cluster closely together, leaving little spread for SNR to predict decision accuracy from. It is a consequence of the limited diversity of the training corpora being compared. With a more diverse set of training mixtures, including some that contain substantial multilingual data and some that do not, we can expect the signal to be higher and the SNR framework to be better positioned to discriminate between reliable and unreliable evaluation sets.

7.4 Summary

Evaluating models on raw multilingual corpora using rolling BPB yields consistently high decision accuracy and a positive SNR vs. decision accuracy correlation, both of which represent improvements over the accuracy-based multilingual downstream task results. The low signal observed across BPB evaluation sets reflects the limited diversity of the DataDecide training mixtures, which are all variations of English web crawl data and therefore produce similar BPB scores on multilingual corpora. Extending this analysis to a more diverse set of training mixtures, including models trained on genuinely multilingual data, would likely produce higher signal and a stronger SNR vs. decision accuracy correlation, making the framework more informative in this setting.

8 Discussion

8.1 Summary of Findings

This work set out to answer whether the Signal and Noise framework for benchmark reliability extends to multilingual settings. Our results across four experiments lead to the following conclusions.

First, we confirm that the framework reproduces reliably on English benchmarks, and that relative dispersion is a stronger signal metric than relative spread, as claimed but not shown in the original paper plots.

Second, we propose k-fold benchmark noise as a practical and effective alternative to checkpoint-to-checkpoint noise. Benchmark noise correlates strongly with checkpoint-to-checkpoint noise ($R = 0.714$, $R^2 = 0.509$) and yields a higher correlation with decision accuracy ($R = 0.808$, $R^2 = 0.653$) than the original noise metric. Crucially, it requires no access to intermediate training checkpoints, making the framework applicable to any model family for which per-question evaluation results are available.

Third, the framework weakens substantially in the multilingual setting when evaluated with English-first models. The SNR vs. decision accuracy correlation drops from $R = 0.594$ on English tasks to $R = 0.045$ on multilingual tasks. After removing the three benchmarks covering languages most underrepresented in the training data, the correlation recovers to $R = 0.293$, suggesting that the framework retains some predictive power when models have at least minimal competence in the evaluated languages.

Fourth, adopting BPB on large multilingual text corpora provides a much stronger SNR vs. decision accuracy correlation compared to using accuracy on downstream tasks, particularly for languages where structured task performance is near-random.

8.2 Limitations

English-first models only. Our experiments use exclusively the DataDecide model suite, which is pre-trained on English-centric corpora. This is a fundamental limitation: the weak results we observe on multilingual tasks are at least partly a consequence of evaluating models that were never meaningfully exposed to the languages being tested. We cannot therefore conclude that the Signal and Noise framework fails for multilingual settings in general — only that it fails when the proxy models lack competence in the evaluated languages. Extending the analysis to genuinely multilingual model families, trained on data that includes meaningful representation of all evaluated languages, is necessary before drawing broader conclusions.

Fold size and statistical power. The k-fold benchmark noise computation partitions each benchmark into 5 folds, which substantially reduces the number of questions per fold. Card et al. [5] provides an explicit sample size formula showing that detecting a performance difference of 3 percentage points between two models at 80% power and 5% significance level requires approximately 1,000 questions. Our fold sizes range from 57 items (`mmlu`, 285 questions divided by 5) to 654 items (`boolq`, 3,270 questions divided by 5). For `mmlu` in particular, each fold contains far fewer questions than Miller's formula would recommend for reliable pairwise comparisons. The strong correlations we observe suggest that this does not prevent benchmark noise from being a meaningful estimate of instability in practice, but it remains a theoretical concern, especially for small benchmarks. Future work could investigate the minimum benchmark size required for k-fold noise to be a reliable proxy, or explore alternative splitting strategies that preserve larger fold sizes.

Training mixture variation only. The DataDecide models vary only in their training data mixture and random seed. They do not vary in architecture, tokeniser, activation function, or other significant hyperparameters. The Signal and Noise framework is motivated in part by the goal of selecting the best model design before a large-scale training run, which implicitly includes architectural choices. We could extend this framework to comparisons across architectures, rather than only across data mixtures, in order to verify this claim.

Inverse scaling tasks. Our multilingual task set includes `truthfulqa-multi`, for which the underlying assumption of the SNR framework — that larger models are better — does not hold. Inverse scaling on this task is a documented phenomenon [12] and suggests that the scaling laws could be adapted to tasks where the relationship between model size and performance is non-monotonic.

8.3 Future Work

Multilingual models. The most important extension of this work is to replicate the multilingual experiments using model families that are genuinely trained on multilingual data. This would allow us to disentangle the effect of model competence from the properties of the benchmarks themselves, and to test whether the Signal and Noise framework provides reliable predictions in settings where the models are actually capable of performing the evaluated tasks.

Architecture variation. The original Signal and Noise paper, and our replication, vary only training data mixture and random seed. A natural extension is to include models with meaningful architectural variation — different tokenisers, activation functions, attention mechanisms, or numbers of layers — to test whether SNR-based predictions generalise to architecture selection decisions. This would make the framework more directly useful for the full range of design choices researchers face before committing to a large-scale training run.

Identifying high-SNR subsets of multilingual tasks. For multilingual benchmarks where small model performance is near-random on many questions, it may be possible to identify a subset of questions for which small model performance is nonetheless indicative of large model performance. Such a subset would act as an aptitude test: rather than testing whether the model already knows the answer, it would test whether the model shows enough promise to be a reliable proxy. Developing methods to identify such subsets automatically would make the framework more applicable in low-resource language settings.

Scaling in language coverage. A further direction is to investigate how SNR behaves as the number of languages in both training data and evaluation sets increases. One could train models on progressively larger multilingual mixtures — starting from a few languages per script and adding more — and track how SNR evolves as a function of language coverage. This would provide a principled way to assess when a multilingual model has acquired sufficient competence for small-scale evaluations to be informative.

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A Appendix

A.1 Training Data Mixtures

The table below describes the composition of each DataDecide training mixture used across our experiments.

Data Mix	Description	Languages	Type
DCLM-baseline	Core high-quality filtered web crawl from the DataComp-LM benchmark.	English	Web (General, Code, Math)
c4	Colossal Clean Crawled Corpus; a classic filtered version of Common Crawl.	English	General Web
dclm-qc-7p-fw2	DCLM baseline refined using a FineWeb-based quality classifier at the 7th percentile threshold.	English	Web
dclm-qc-7p-fw3	Similar to fw2, using a different iteration or weighting of the FineWeb-based filter.	English	Web
dclm-fw-top10	DCLM baseline filtered to match the top 10% of FineWeb quality scores.	English	High-quality Web
dclm-fw-top3	DCLM baseline filtered to match the top 3% of FineWeb quality scores.	English	Elite-quality Web
dolma-v1-6-and-sources	AI2's Dolma v1.6, consisting of web content, academic papers, books, and code.	English	Mixed (Web, Books, Code, STEM)
dolma1_7	Updated v1.7 release of the Dolma dataset.	English	Mixed (Web, Books, Code)
dolma1.7-25p / DCLM-75p	Proportional mix: 25% Dolma 1.7 and 75% DCLM-baseline.	English	Web / Mixed
dolma1.7-50p / DCLM-50p	Balanced 50/50 mix of Dolma 1.7 and DCLM-baseline.	English	Web / Mixed
dolma1.7-75p / DCLM-25p	Proportional mix: 75% Dolma 1.7 and 25% DCLM-baseline.	English	Web / Mixed
falcon	RefinedWeb dataset used to train the original Falcon models; heavy deduplication.	English	General Web
falcon-and-cc	Combination of the Falcon (RefinedWeb) dataset and raw Common Crawl.	English	General Web
falcon-and-cc-eli5-oh-top10p	Falcon+CC filtered using quality signals from ELI5 and OpenHermes (top 10%).	English	Reasoning / Knowledge
falcon-and-cc-eli5-oh-top20p	Falcon+CC filtered using quality signals from ELI5 and OpenHermes (top 20%).	English	Reasoning / Knowledge
falcon-and-cc-og-eli5-oh-top10p	Original Falcon+CC filtered by ELI5/OpenHermes signals.	English	Reasoning / Knowledge

falcon-and-cc-tulu-qc-top10	Falcon+CC filtered using a quality classifier trained on Tulu instruction data.	English	Reasoning / Instructions
fineweb-edu-dedup	HuggingFace FineWeb-edu; filtered for educational and informative content.	English	Educational / Academic
no-code	DCLM-baseline with all programming code content removed.	English	General Web (no code)
no-flan	DCLM-baseline with instruction-tuning data (FLAN) removed.	English	General Web
no-math-no-code	DCLM-baseline with mathematical notation and programming code removed.	English	General Web (no STEM)
no-reddit	DCLM-baseline with all Reddit-sourced conversational data removed.	English	General Web (no forums)
pos-eli5-oh-neg-dclm-top10p	Complex filter using ELI5/OpenHermes as positive signals and RefinedWeb as negative (top 10%).	English	High-reasoning Web
pos-eli5-oh-neg-dclm-top20p	Same as above at top 20%.	English	High-reasoning Web
prox-fineweb-pro	Data selected to be statistically similar to the FineWeb Professional subset.	English	High-quality Web

Table 1: Description of DataDecide training mixtures used across all experiments.

A.2 Model Configurations per Experiment

Experiment	Data Mixes	Size Range	Seeds	Checkpoints
Reproduction	Full DataDecide suite (25 mixes)	4M – 1B	14, 15, 6198	Last 30% of training
Benchmark Noise	DCLM-baseline, falcon-and-cc-tulu-qc-top10, dolma1.7-75p-DCLM-25p, dolma1.7-50p-DCLM-50p, dolma1.7-25p-DCLM-75p, dclm-qc-7p-fw2, dclm-qc-7p-fw3, dclm-fw-top10, dclm-fw-top3, dolma-v1-6-and-sources	20M, 150M, 530M, 1B	6198	Final checkpoint only
Multilingual	DCLM-baseline, dclm-25p-dolma1.7-75p, dclm-50p-dolma1.7-50p, dclm-75p-dolma1.7-25p, dclm-qc-10p, dclm-qc-20p, dclm-qc-7p-fw2, dclm-qc-7p-fw3, dclm-qc-fw-10p, dclm-qc-fw-3p, dolma1_6plus	4M – 1B	Default revision	Default revision

Table 2: Model configurations used across the three main experiments.

A.3 Reproduction Benchmark Details

Task	Format	Domain	Instances
arc_challenge	MCQ	Grade-school science	299
arc_easy	MCQ	Grade-school science	570
csqa	MCQ	Commonsense knowledge	12,247
hellaswag	MCQ	Everyday activities	
mmlu	MCQ	57 academic subjects	285
openbookqa	MCQ	Elementary science	500
piqa	MCQ	Physical commonsense	1,838
socialiqa	MCQ	Social interactions	1,954
winogrande	Binary choice	Commonsense reasoning	1,267
boolq	Binary choice	Reading comprehension	3,270

Table 3: Reproduction benchmark details. The tasks are in English only and evaluated using accuracy metric.

A.4 Multilingual Task Details

A.4.1 Multilingual downstream tasks

Task	Description	Languages
Non English-only tasks		
bangla_mmlu	MMLU-style benchmark for Bengali (Bangla), evaluating knowledge and NLP capabilities.	Bengali
bertaqa	Local Basque cultural trivia QA in English and Basque.	Basque, English
belebele	Multiple-choice reading comprehension spanning 122 language variants.	122 languages
blimp_nl	Dutch version of BLiMP, testing grammatical phenomena.	Dutch
cabbq	Adaptation of BBQ to Spanish stereotypes prevalent in Spain.	Spanish
click	Benchmark for cultural and linguistic intelligence in Korean.	Korean
crows_pairs	Social bias benchmark comparing sentence pairs across demographic groups.	French, English
darijahellaswag	HellaSwag adapted to Moroccan Darija.	Moroccan Darija
egyhellaswag	HellaSwag adapted to Egyptian Arabic (Masri).	Egyptian Arabic

eus_proficiency	Basque language proficiency tasks across various topics.	Basque
eus_reading	Basque reading comprehension tasks.	Basque
eus_trivia	Trivia and knowledge testing in Basque.	Basque
truthfulqa-multi	Multilingual TruthfulQA evaluating model truthfulness across languages.	Arabic, Chinese, French, German, Spanish, and others
turblimp_core	Turkish version of BLiMP, testing grammatical phenomena.	Turkish
turkishmmlu	Turkish MMLU with questions from 9 Turkish high-school subjects.	Turkish
xcopa	Causal commonsense reasoning across 11 languages.	Estonian, Haitian Creole, Indonesian, Italian, Quechua, Swahili, Chinese, Tamil, Thai, Turkish, Vietnamese
xnli_eu	Natural language inference in Basque.	Basque
xstorycloze	Story ending prediction across 10 languages.	Arabic, English, Spanish, Basque, Indonesian, Burmese, Russian, Swahili, Chinese, Telugu
xwinograd	Multilingual Winograd schema challenge across 6 languages.	English, French, Japanese, Portuguese, Russian, Chinese
English-only tasks		
anli	Adversarial NLI with hard entailment cases.	English
asdiv	Arithmetic reasoning from elementary math word problems.	English
commonsense_qa	Multiple-choice commonsense knowledge QA.	English
groundcocoa	Conditional and compositional reasoning benchmark.	English
hellaswag	Commonsense NLI sentence completion.	English
lambada	Predicting the final word of a passage requiring broad discourse context.	English
lambada_cloze	Cloze-style version of LAMBADA.	English
winogrande	Large-scale commonsense pronoun resolution.	English
winogender	Gender bias in pronoun resolution.	English

wmdp	Hazardous knowledge proxy benchmark (biosecurity, cybersecurity, chemical).	English

Table 4: Full list of multilingual and English-only tasks used in the multilingual experiment.

A.4.2 Multilingual BPB evaluation sets

A.5 BPB Evaluation Set Details

Task Group	**Languages Covered**
Flores+ subsets	
`flores_plus_bpb_1`	Acehnese (Arab, Latn), Mesopotamian Arabic, Ta'izzi-Adeni Arabic, Tunisian Arabic, Afrikaans, Tosk Albanian, Amharic, North Levantine Arabic (2 varieties), Modern Standard Arabic (Arab, Latn), Aragonese, Najdi Arabic, Moroccan Arabic, Egyptian Arabic, Assamese, Asturian, Awadhi, Aymara, South Azerbaijani, North Azerbaijani
`flores_plus_bpb_2`	Bashkir, Bambara, Balinese, Belarusian, Bemba, Bengali, Bhojpuri, Banjar (Arab, Latn), Tibetan, Bosnian, Bodo, Buginese, Bulgarian, Catalan (incl. Valencian), Cebuano, Czech, Chuvash, Chokwe, Central Kurdish, Mandarin Chinese (Simplified, Traditional)
`flores_plus_bpb_3`	Crimean Tatar, Welsh, Danish, Dargwa, German, Dogri, Dinka, Dyula, Dzongkha, Estonian, Greek, English, Esperanto, Basque, Ewe, Faroese, Fijian, Filipino, Finnish, Fon, French, Friulian, Nigerian Fulfulde
`flores_plus_bpb_4`	West Central Oromo, Scottish Gaelic, Irish, Galician, Goan Konkani, Paraguayan Guaraní, Gujarati, Haitian Creole, Hausa, Hebrew, Hindi, Chhattisgarhi, Croatian, Hungarian, Armenian, Igbo, Ilocano, Indonesian, Icelandic, Italian, Javanese, Japanese, Kara-Kalpak
`flores_plus_bpb_5`	Kabyle, Kachin, Kamba, Kannada, Kashmiri (Arab, Deva), Georgian, Kazakh, Kabiyè, Kabuverdianu, Mongolian (Cyrl, Mong), Khmer, Kikuyu, Kinyarwanda, Kyrgyz, Kimbundu, Kurmanji Kurdish, Central Kanuri (Arab, Latn), Korean, Lao
`flores_plus_bpb_6`	Ligurian, Limburgish, Lingala, Lithuanian, Ladin (2 varieties), Lombard, Latgalian, Luxembourgish, Luba-Kasai, Ganda, Luo, Mizo, Latvian, Magahi, Maithili, Malayalam, Marathi, Morisyen, Eastern Mari, Minangkabau (Arab, Latn), Macedonian
`flores_plus_bpb_7`	Maltese, Meitei (Beng, Mtei), Mossi, Māori, Burmese, Erzya, Dutch, Norwegian Nynorsk, Norwegian Bokmål (2 varieties), Nepali, N'Ko, Northern Sotho, Nuer, Chichewa, Occitan (incl. Aranese), Odia, Pangasinan, Punjabi, Papiamento, Southern Pashto
`flores_plus_bpb_8`	Western Persian, Plateau Malagasy, Polish, Portuguese, Dari, Ayacucho Quechua, Romanian, Rundi, Russian, Sango, Sanskrit, Santali, Sicilian, Shan, Sinhala, Slovak, Slovenian, Samoan, Shona, Sindhi (Arab, Deva), Somali, Southern Sotho

`flores_plus_bpb_9`	Spanish, Sardinian, Serbian, Swati, Sundanese, Swedish, Swahili, Silesian, Tamil, Tamasheq (Latn, Tfng), Tatar, Telugu, Tajik, Thai, Tigrinya, Tok Pisin, Tswana, Tsonga, Turkmen, Tumbuka, Turkish
`flores_plus_bpb_10`	Twi (2 varieties), Tuvan, Uyghur, Ukrainian, Umbundu, Urdu, Uzbek (Latn, Arab), Venetian, Vietnamese, Makhuwa-Meetto, Waray, Wolof, Wu Chinese, Xhosa, Yiddish, Yoruba, Yue Chinese, Standard Moroccan Tamazight, Malay, Zulu
Wiki40B subsets	
`wiki40b_bpb_1`	Arabic, Bulgarian, Catalan, Czech, Danish, German, Greek, English
`wiki40b_bpb_2`	Spanish, Estonian, Persian, Finnish, French, Hebrew, Hindi, Croatian
`wiki40b_bpb_3`	Hungarian, Indonesian, Italian, Japanese, Korean, Lithuanian, Latvian, Malay
`wiki40b_bpb_4`	Dutch, Norwegian, Polish, Portuguese, Romanian, Russian, Slovak, Slovenian
`wiki40b_bpb_5`	Serbian, Swedish, Thai, Filipino, Turkish, Ukrainian, Vietnamese, Chinese (Simplified), Chinese (Traditional)

Table 5: Language coverage of BPB evaluation set groups. Each Flores+ group covers approximately 22–23 languages spanning diverse language families and scripts. Each Wiki40B group covers 8–9 languages. Evaluation instance counts per language are inherited from the original dataset test splits.

A.6 Additional Plots for Benchmark Noise Experiment

The following figures present the full set of results for the benchmark noise experiment, varying both the models used to compute benchmark noise (150M only vs. all sizes from 20M to 1B) and the model sizes used to compute signal and decision accuracy.

Benchmark Noise Computed from 150M Models Only

Benchmark Noise Computed from All Models (20M to 1B)

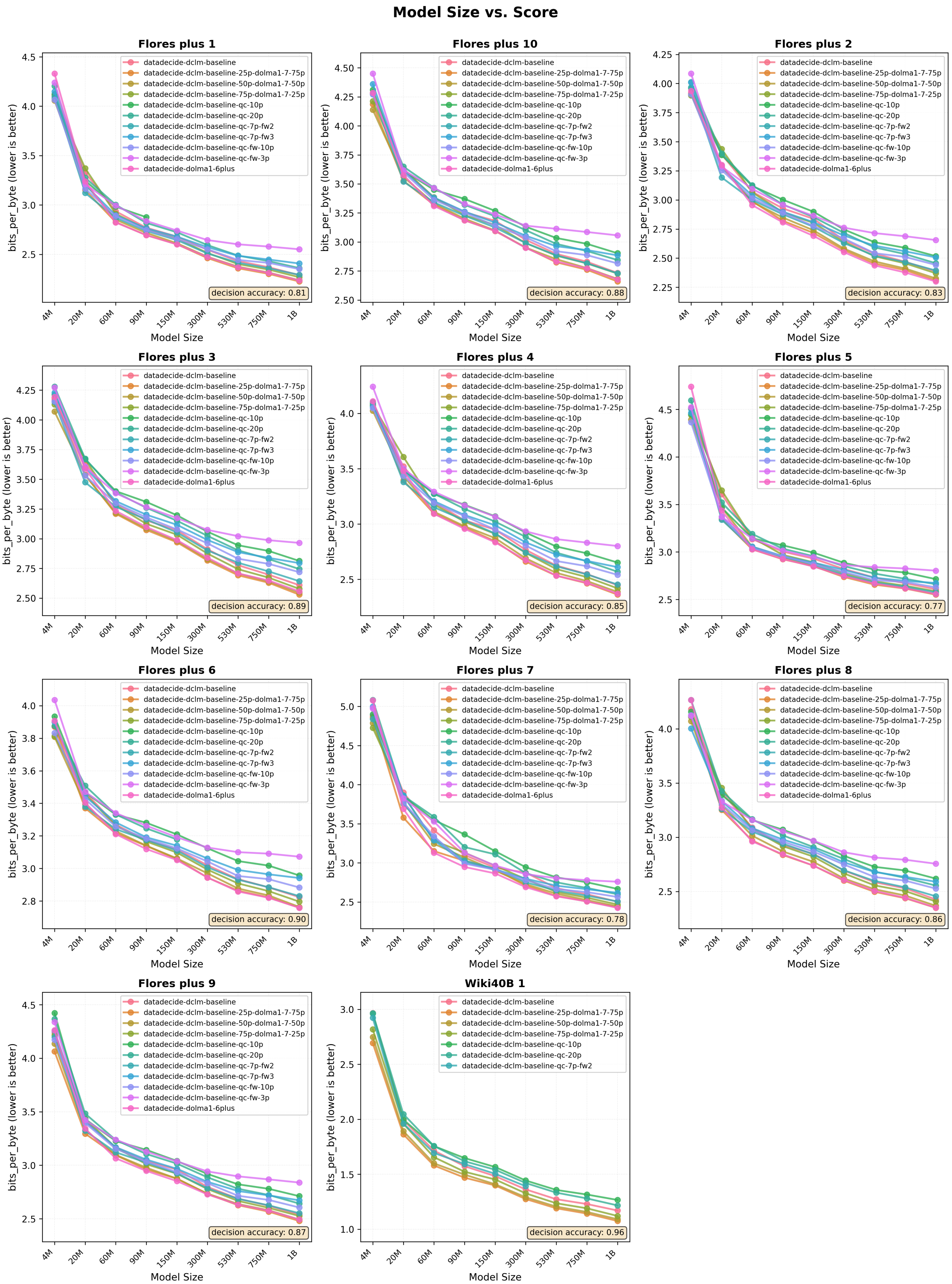

Figure 8: Model BPB as a function of model size across BPB evaluation sets, for all 11 DataDecide training corpora. Lower BPB is better. Decision accuracy is consistently high across all subsets (0.77–0.96), indicating that small model rankings reliably agree with 1B model rankings even on multilingual raw text corpora.

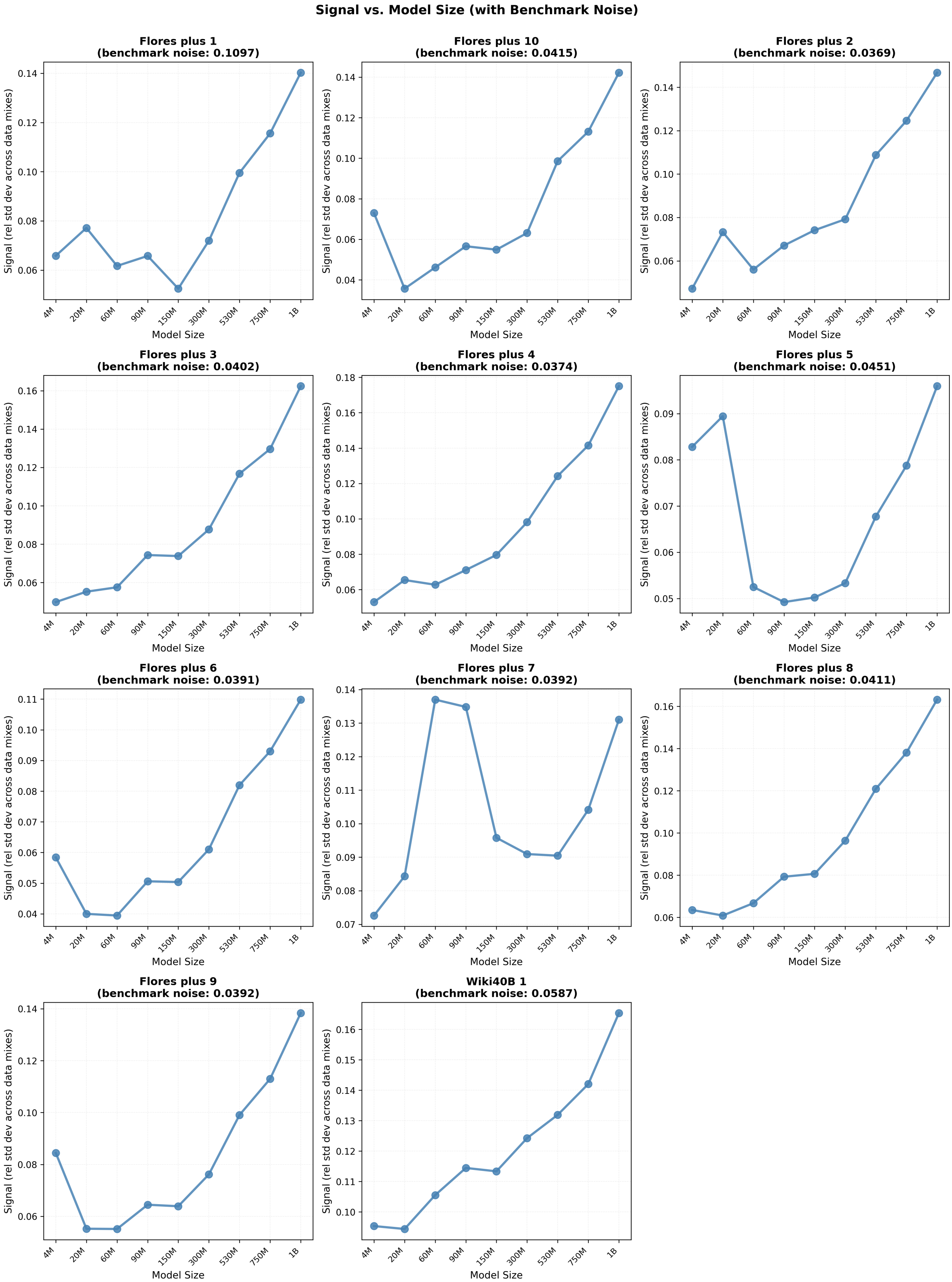

Figure 9: Signal as a function of model size across BPB evaluation sets. Signal values are consistently low across all subsets, reflecting the fact that English-first training mixtures produce similar BPB scores on multilingual corpora. Benchmark noise values are shown in the subplot titles.

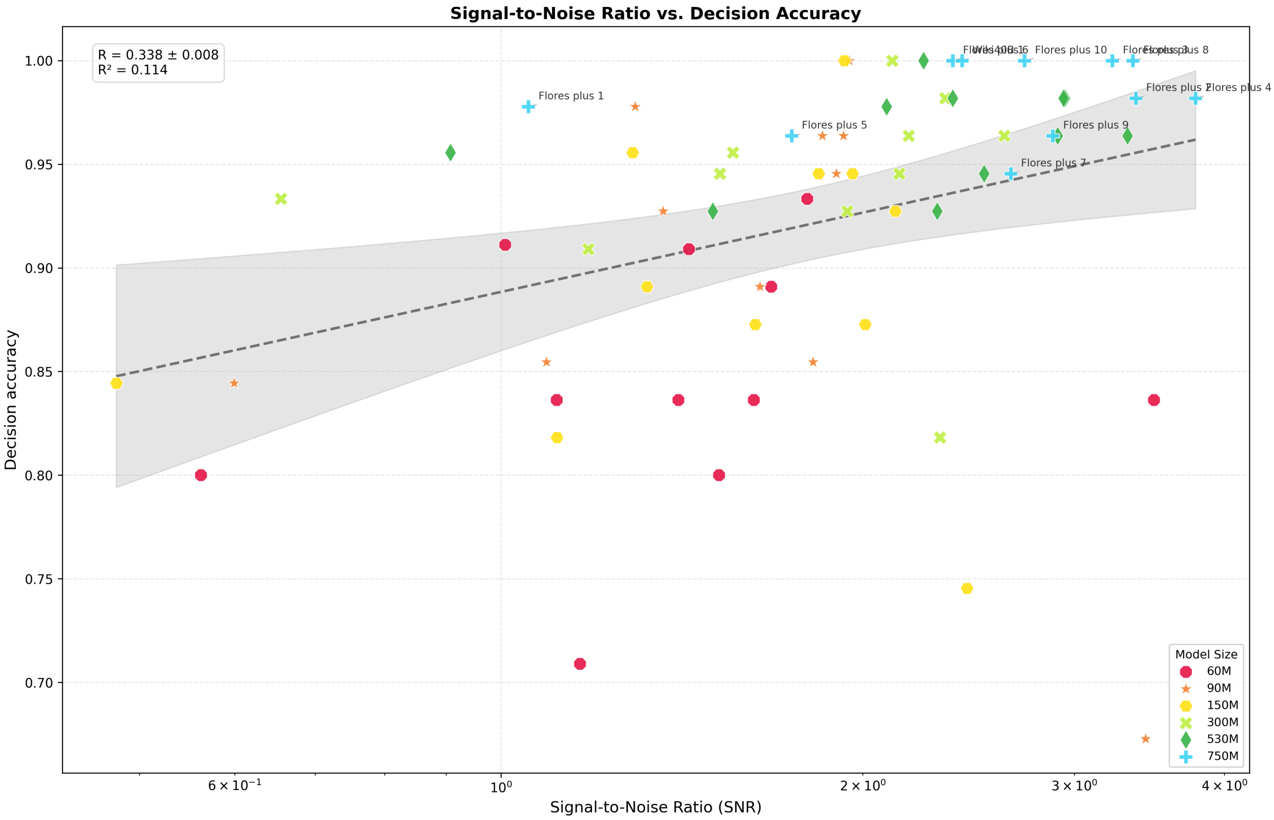

Figure 10: SNR vs. decision accuracy on BPB evaluation sets ($R = 0.307$, $R^2 = 0.094$). Despite the models being English-first and signal being low, the correlation is substantially higher than on non-English downstream tasks ($R = 0.045$), suggesting that rolling BPB on raw corpora provides a more reliable evaluation signal in the multilingual setting.

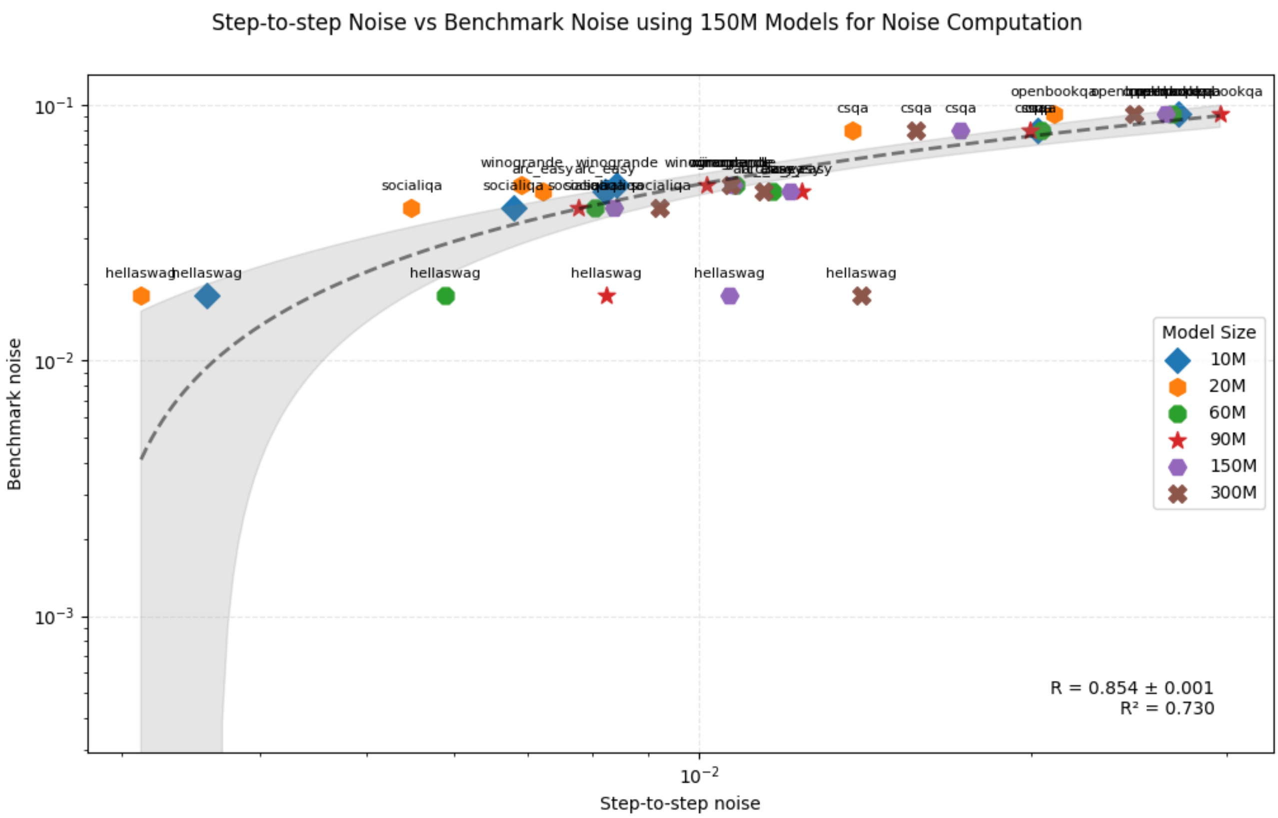

Figure 11: Checkpoint-to-checkpoint noise vs. benchmark noise (computed from 150M models), using models of sizes 10M to 300M for checkpoint noise. Strong correlation confirms that benchmark noise is a reliable substitute for checkpoint-to-checkpoint noise at small model scales.

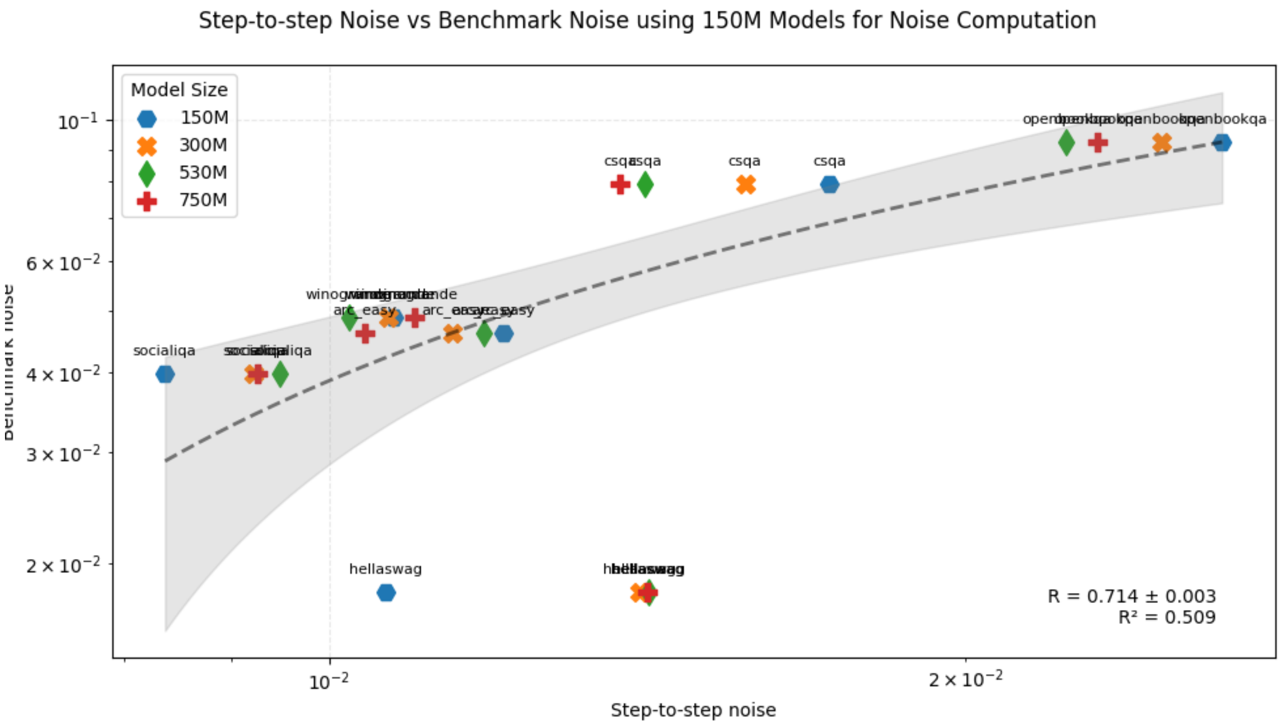

Figure 12: Checkpoint-to-checkpoint noise vs. benchmark noise (computed from 150M models), using models of sizes 150M to 750M for checkpoint noise. The correlation holds across larger model sizes, confirming robustness of the result.

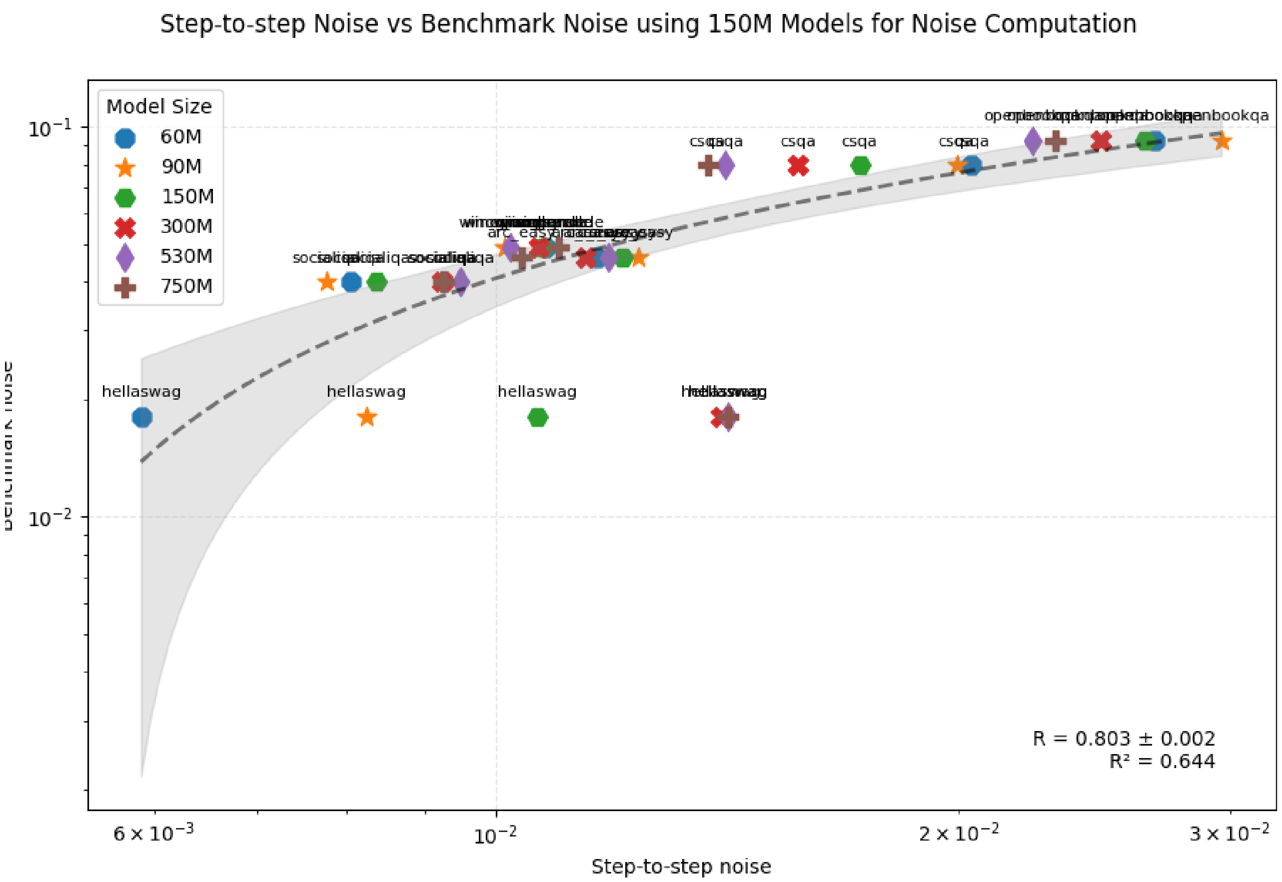

Figure 13: Checkpoint-to-checkpoint noise vs. benchmark noise (computed from 150M models), using models of sizes 60M to 750M for checkpoint noise.

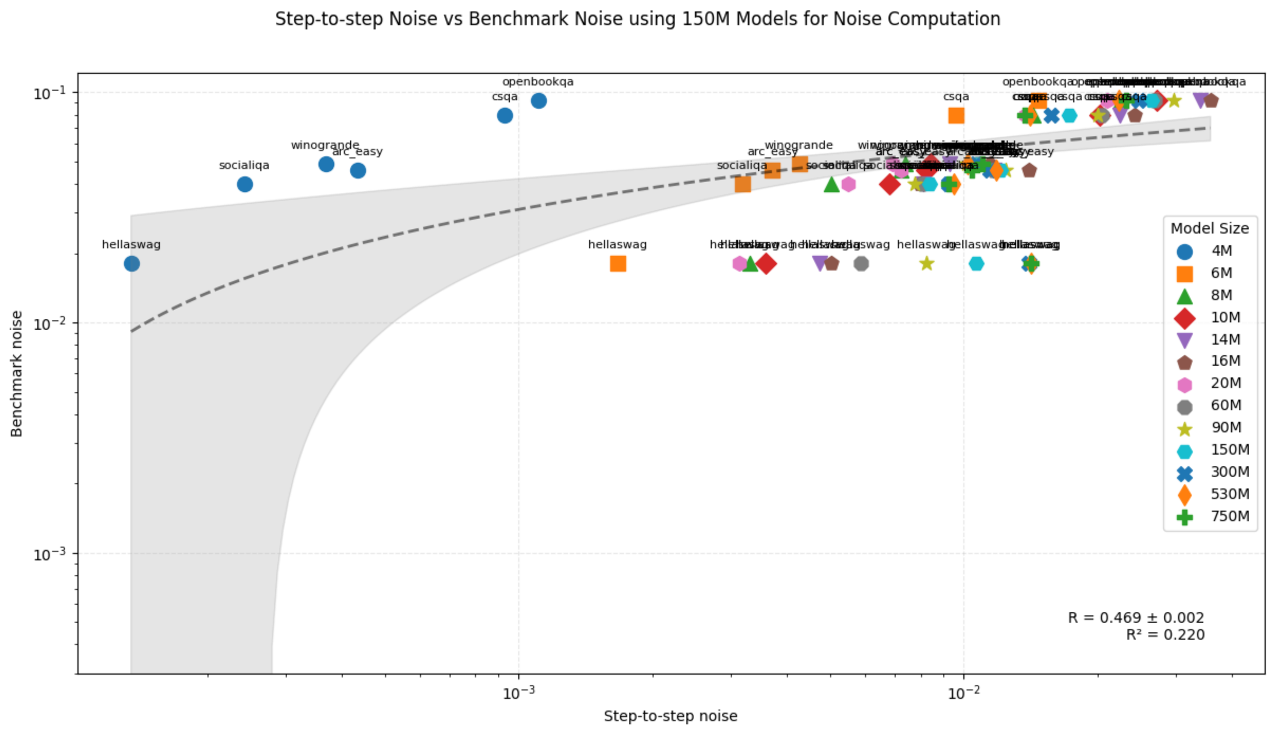

Figure 14: Checkpoint-to-checkpoint noise vs. benchmark noise (computed from 150M models), using all available model sizes for checkpoint noise.

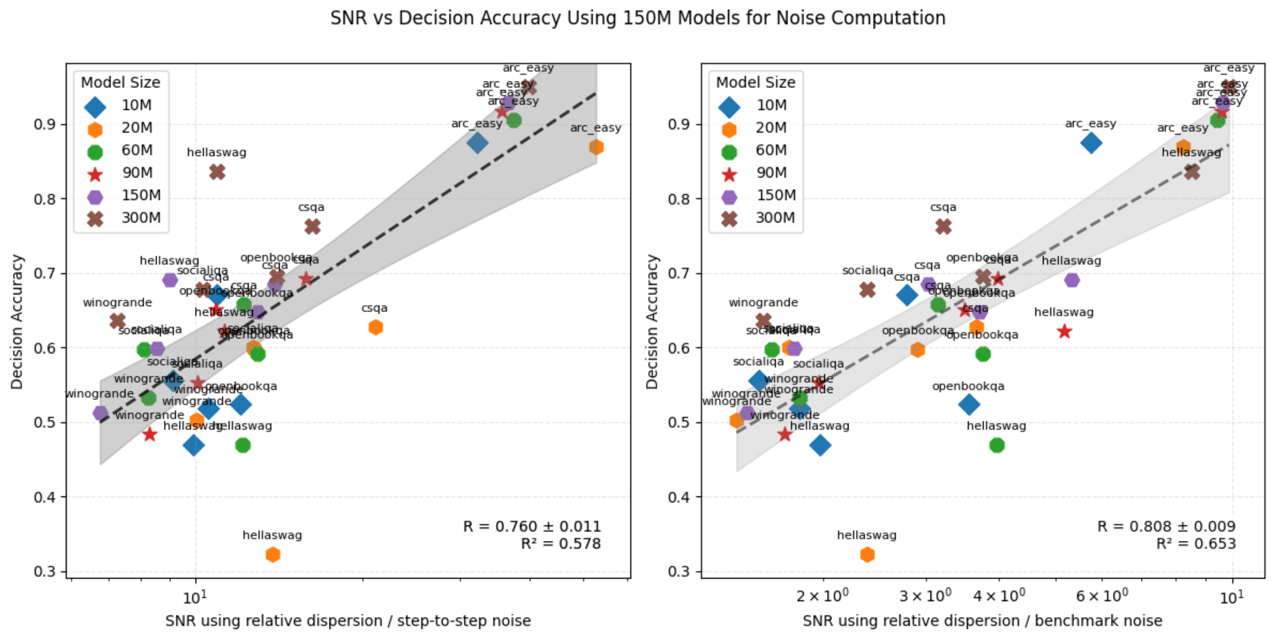

Figure 15: SNR vs. decision accuracy (benchmark noise computed from 150M models), using models of sizes 10M to 300M. This is the main result presented in Section 5.

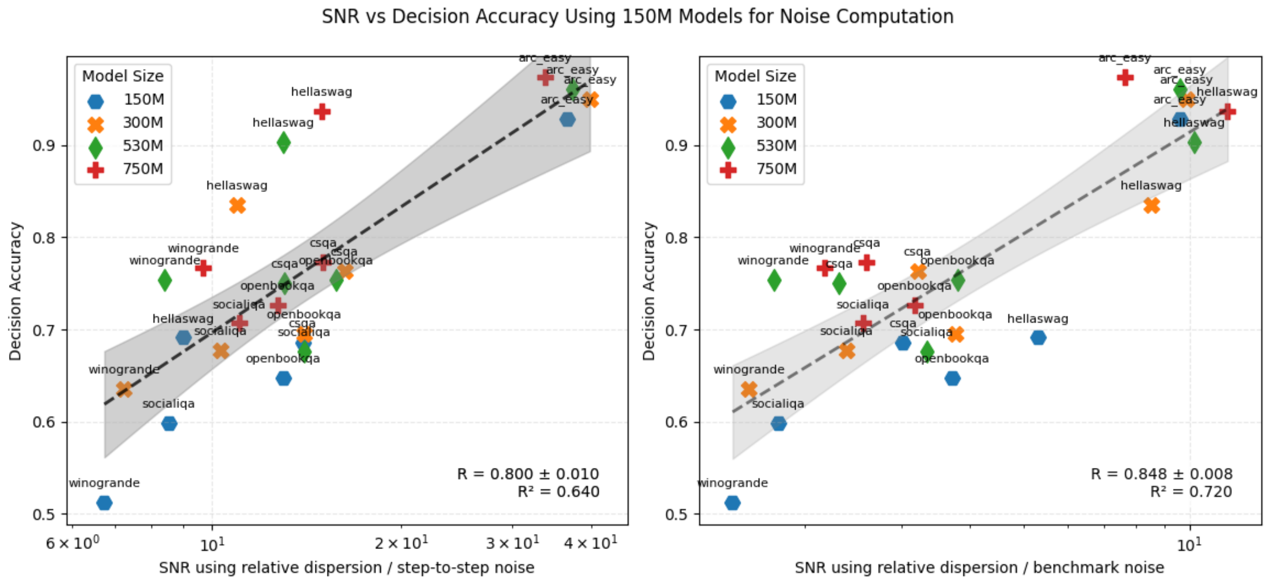

Figure 16: SNR vs. decision accuracy (benchmark noise computed from 150M models), using models of sizes 150M to 750M.

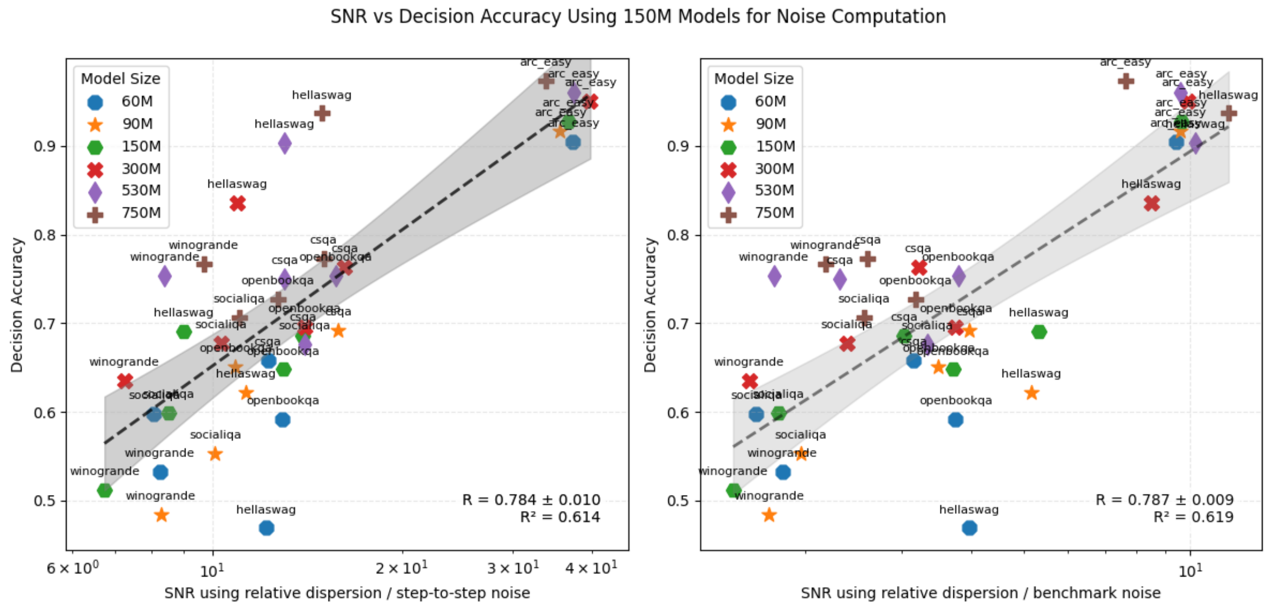

Figure 17: SNR vs. decision accuracy (benchmark noise computed from 150M models), using models of sizes 60M to 750M.

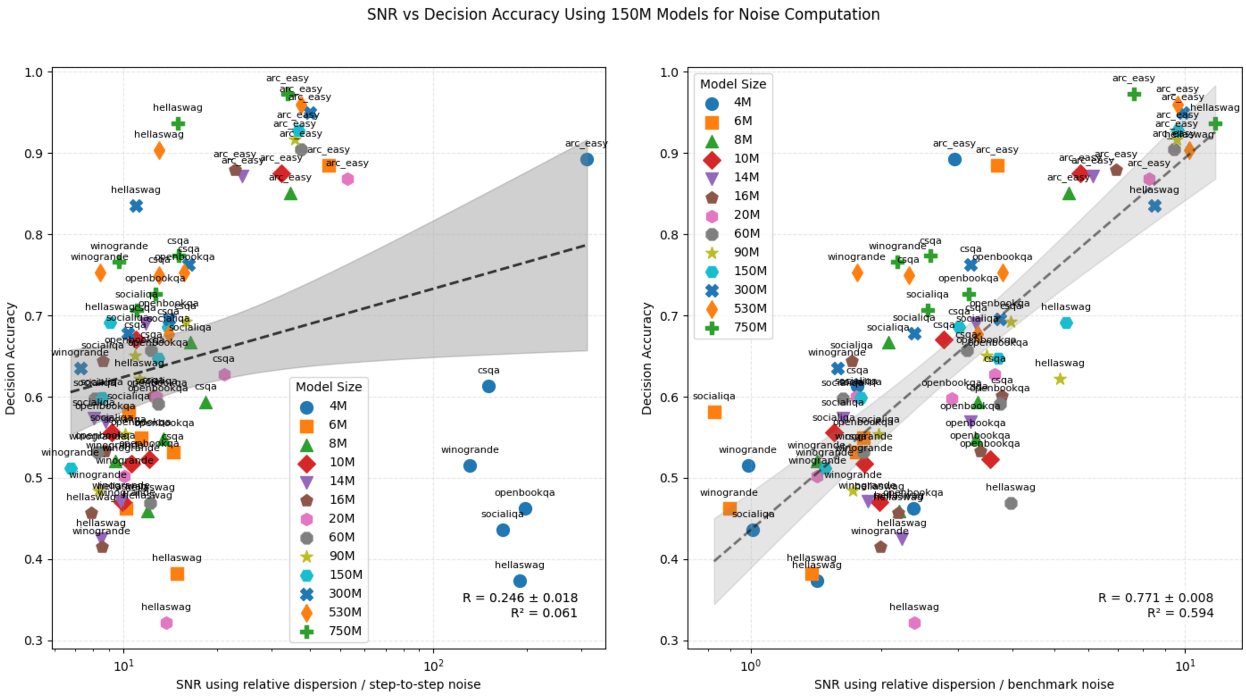

Figure 18: SNR vs. decision accuracy (benchmark noise computed from 150M models), using all available model sizes.

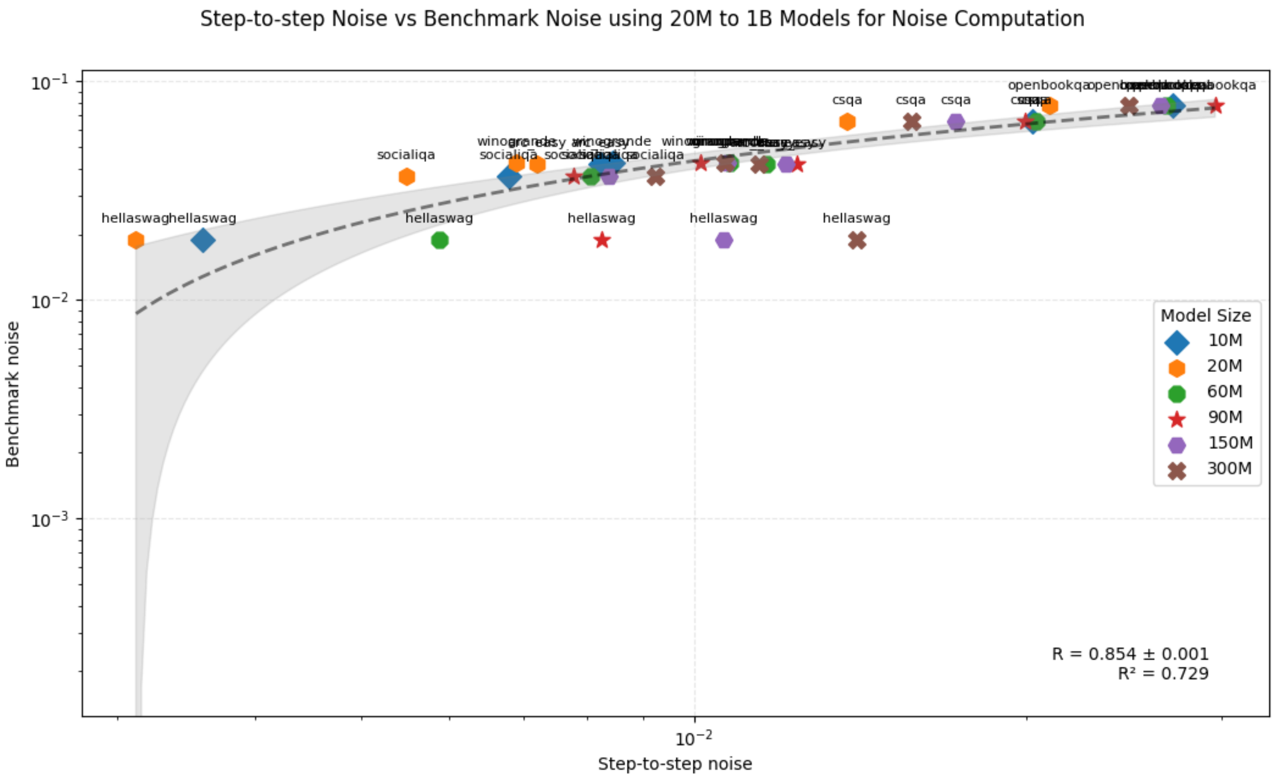

Figure 19: Checkpoint-to-checkpoint noise vs. benchmark noise (computed from all models, 20M to 1B), using models of sizes 10M to 300M for checkpoint noise. Confirms that the correlation between the two noise metrics holds when benchmark noise is computed from a broader set of model sizes.

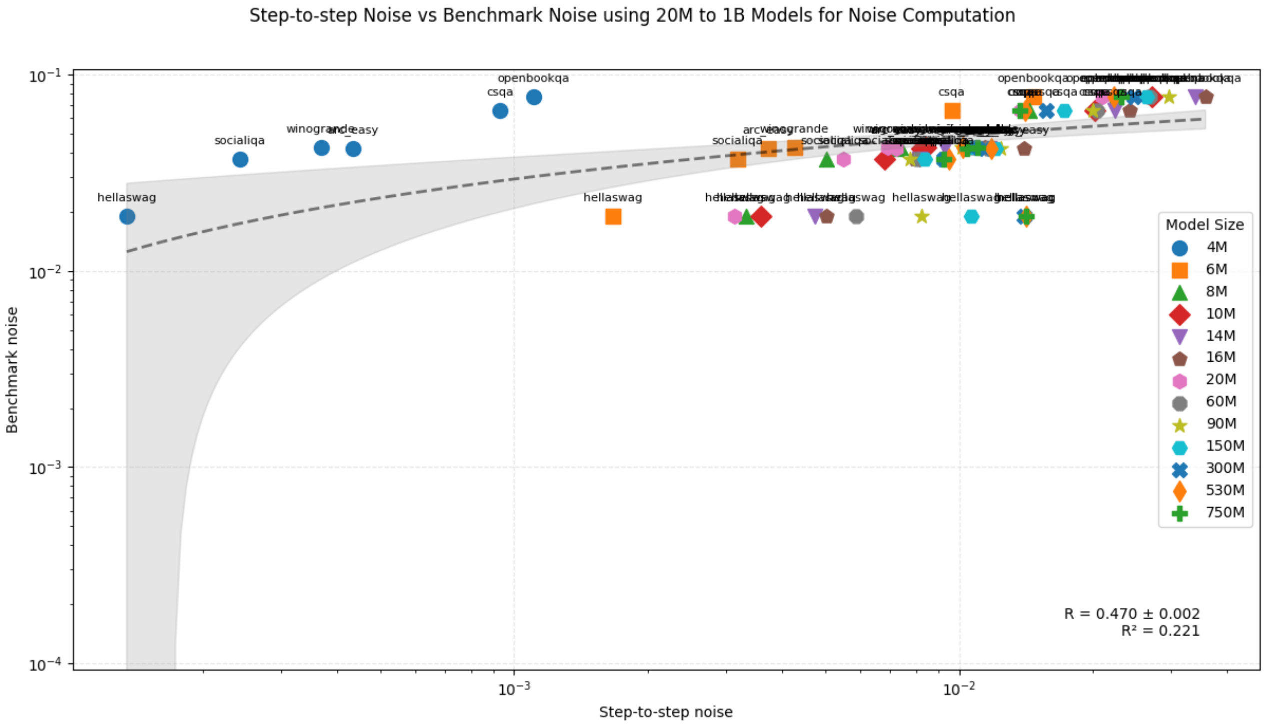

Figure 20: Checkpoint-to-checkpoint noise vs. benchmark noise (computed from all models, 20M to 1B), using all available model sizes for checkpoint noise.

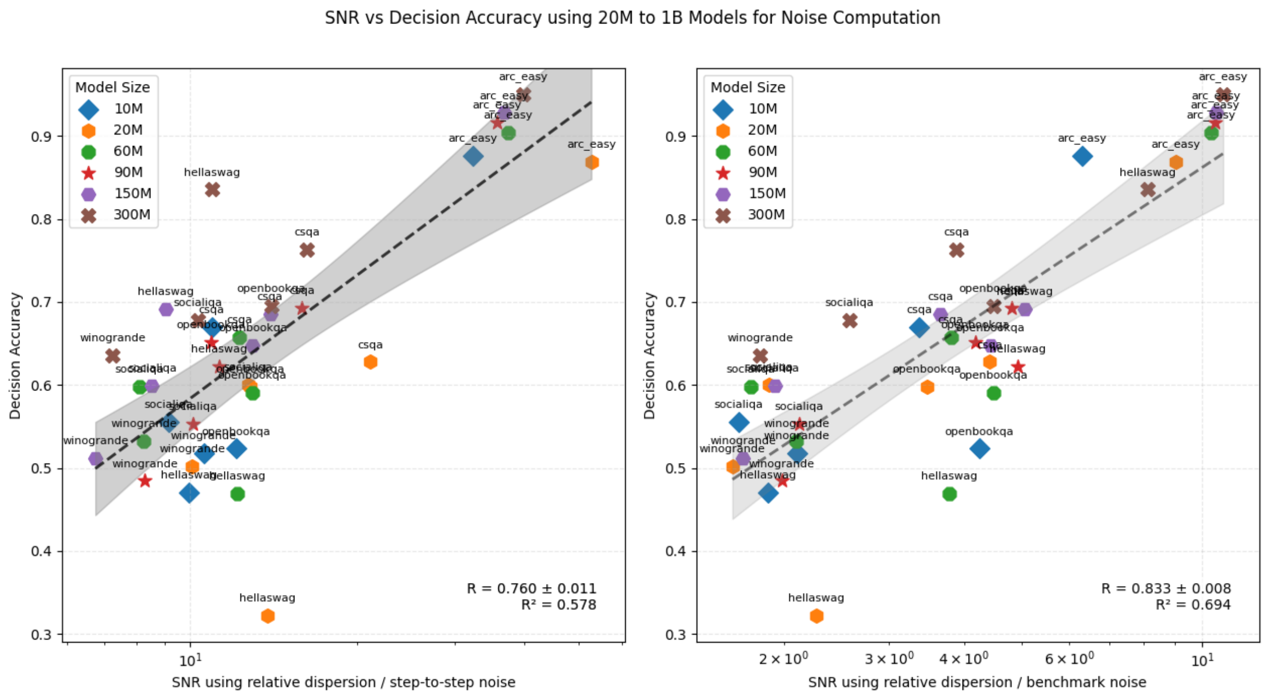

Figure 21: SNR vs. decision accuracy (benchmark noise computed from all models, 20M to 1B), using models of sizes 10M to 300M. The strong correlation confirms that the improvement in SNR predictive power is not specific to using 150M models for noise computation.

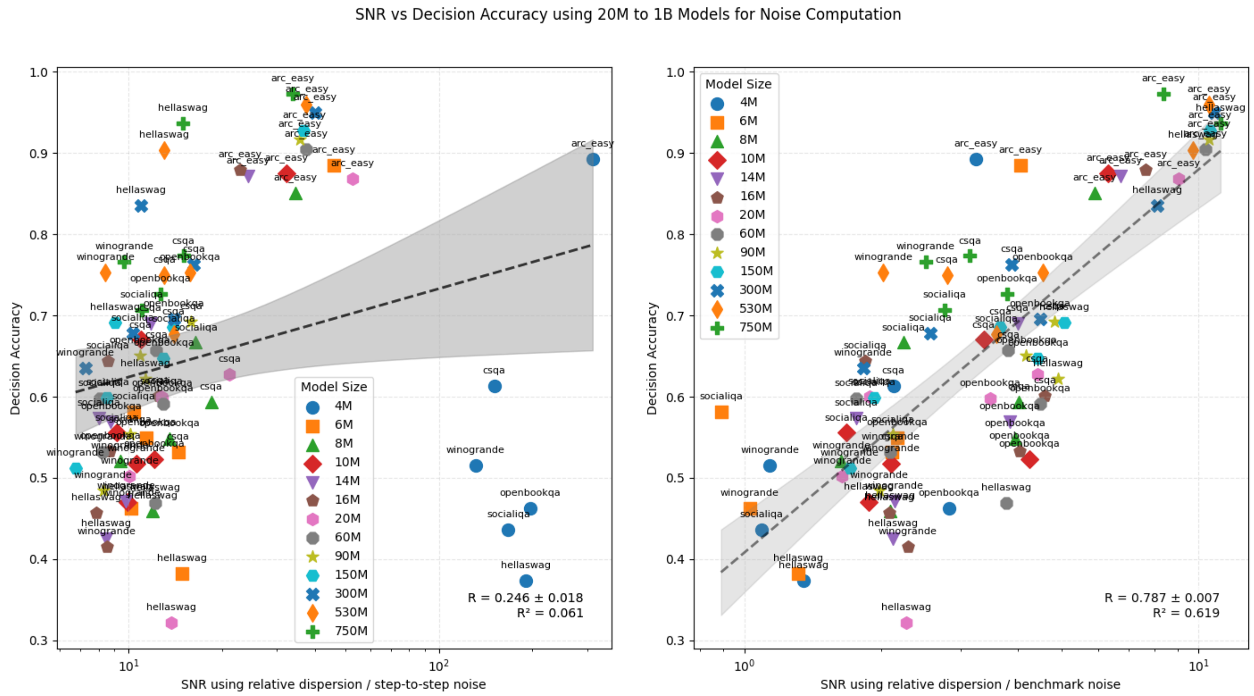

Figure 22: SNR vs. decision accuracy (benchmark noise computed from all models, 20M to 1B), using all available model sizes.